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Measurements of the Higgs boson production, fiducial and differential cross-sections in the four lepton decay channel using 164 fb^{-1} of data collected at $\sqrt{s} = 13.6 \text{ TeV}$ with the ATLAS detector

The ATLAS Collaboration

Inclusive, differential and production-mode cross-section measurements of the Higgs boson are performed in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel. The analysis uses proton–proton collision data produced at the Large Hadron Collider at a centre-of-mass energy of $\sqrt{s} = 13.6 \text{ TeV}$ and recorded with the ATLAS detector, corresponding to an integrated luminosity of 164 fb^{-1} . The inclusive fiducial cross-section for the $H \rightarrow ZZ^* \rightarrow 4\ell$ process is measured to be $\sigma_{\text{fid}} = 3.65^{+0.35}_{-0.33} \text{ fb}$, in agreement with the Standard Model prediction of $\sigma_{\text{fid}}^{\text{SM}} = 3.68 \pm 0.17 \text{ fb}$. Differential cross-sections are measured as a function of key kinematic observables of the Higgs boson and the four-lepton final state. Cross-sections are measured for the main production-modes in several exclusive regions of the Higgs boson production phase space and combined to measure an overall Higgs boson signal strength, defined as the measured cross-section normalised to the SM prediction, of $\mu = 0.99 \pm 0.13$. The results are interpreted in terms of modifications of Higgs boson couplings using the κ framework, within the Standard Model Effective Field Theory, and in scenarios probing the Higgs boson self-coupling. All the results are consistent with Standard Model expectations.

1 Introduction

The observation of the Higgs boson by the ATLAS and CMS Collaborations [1, 2] using proton–proton (pp) collisions at the Large Hadron Collider (LHC) marked a major milestone in the understanding of the electroweak (EW) symmetry-breaking mechanism [3–5]. Subsequent studies of the Higgs boson properties have established the observed particle as a scalar boson consistent with the Standard Model (SM) predictions for a Higgs boson with a mass of 125.09 ± 0.24 GeV [6].

The $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel, where $\ell = e, \mu$, provides a particularly clear experimental signature, featuring an excellent signal-to-background ratio and full reconstruction of the final state enabled by the high-precision lepton momentum measurements of the ATLAS detector. Background contributions are small and can be precisely estimated using data-driven techniques. As a result, this channel has played a central role in precision measurements of Higgs boson properties by both the ATLAS and CMS experiments using the full LHC Run 2 integrated luminosity of 140 fb^{-1} . At a centre-of-mass energy of $\sqrt{s} = 13$ TeV, measurements of the Higgs boson production and differential cross-sections [7–13], and derived constraints on couplings, as well as its mass [14–16], tensor structure [17, 18], and width [16, 19], have been performed. More recently, first measurements at $\sqrt{s} = 13.6$ TeV have been reported by ATLAS [20] and CMS [21].

This note presents measurements of Higgs boson production using $H \rightarrow ZZ^* \rightarrow 4\ell$ decay candidates recorded during Run 3 with the ATLAS experiment in pp collisions at $\sqrt{s} = 13.6$ TeV between the 2022 and 2024 data-taking periods, corresponding to an integrated luminosity of 164 fb^{-1} . Inclusive and differential cross-sections for eight observables sensitive to the production and decay kinematics of the Higgs boson, are measured within a well-defined fiducial phase space. In addition, cross-sections per Higgs boson production-mode are extracted using the Simplified Template cross-section (STXS) framework [22, 23].

The cross-sections are determined using binned likelihood fits to the reconstructed four-lepton invariant mass spectrum for the inclusive and differential cross-sections measurement, and are corrected for detector efficiencies and resolution effects through an unfolding procedure. Neural-network-based (NN) discriminant distributions are used for the production-mode cross-sections measurements.

The measured cross-sections are compared with SM predictions and are further interpreted to constrain possible anomalous Higgs boson couplings. In particular, the differential production cross-sections are interpreted in the context of the Standard Model effective field theory (SMEFT), while the cross-sections per production-mode are interpreted in terms of modified Higgs boson couplings to vector bosons. Constraints on the Higgs boson trilinear self-coupling are derived by exploiting both the Higgs boson transverse momentum spectra and the production-mode cross-sections. Furthermore, a statistical combination of Run 2 and Run 3 data is performed in the context of the coupling-modifier interpretation, significantly improving the precision of the constraints.

Compared with previous results in this analysis channel [8, 9, 20], the results presented in this note benefit from additional data collected in 2023 and 2024. A new transformer-based [24] machine learning (ML) algorithm is used to identify the original quark flavour of hadronic jets, providing substantial improvements in background rejection [25] for measuring the top-associated Higgs production-mode. A data-driven fake-factor method [26] is used to estimate backgrounds with non-prompt leptons, such as those arising from Z +jets, $t\bar{t}$, and WZ processes. A new double-differential distribution of the transverse momentum of the four-lepton system as function of the invariant mass of the sub-leading dilepton pair is being measured for the first time, owing to its high sensitivity to possible beyond-the-Standard-Model (BSM) effects in Higgs boson couplings. A new neural-network architecture is also employed to construct discriminants

that enhance the sensitivity to distinguish between the various Higgs boson production-modes. In addition, the azimuthal angle between the decay planes of the two Z bosons, ϕ , was found to improve sensitivity to SM-BSM interference effects and is reinterpreted in the context of SMEFT for the first time in this channel.

The note is organised as follows. Section 2 provides a brief description of the ATLAS detector. The dataset used in the analysis, together with the simulated signal and background samples, is described in Section 3. The event selection for Higgs boson candidates is presented in Section 4, while the estimation of background contributions is discussed in Section 5. An overview of the analysis methodology used to measure inclusive and differential cross-sections in a fiducial phase space, as well as production-mode cross-sections in the STXS framework, is given in Section 6. The experimental and theoretical systematic uncertainties affecting the measurements are summarised in Section 7. The cross-section results are presented in Section 8, and their interpretation is discussed in Section 9. Finally, the conclusions are given in Section 10.

2 ATLAS detector

The ATLAS experiment [27, 28] at the LHC is a multipurpose particle detector with a forward–backward symmetric cylindrical geometry and a near 4π coverage in solid angle.¹ It consists of an inner tracking detector (ID) surrounded by a thin superconducting solenoid providing a 2 T axial magnetic field, electromagnetic and hadronic calorimeters, and a muon spectrometer. The inner tracking detector covers the pseudorapidity range $|\eta| < 2.5$. It consists of silicon pixel, silicon microstrip, and transition radiation tracking (TRT) detectors. Lead/liquid-argon (LAr) sampling calorimeters provide electromagnetic (EM) energy measurements with high granularity within the region $|\eta| < 3.2$. A steel/scintillator-tile hadronic calorimeter covers the central pseudorapidity range ($|\eta| < 1.7$). The endcap and forward regions are instrumented with LAr calorimeters for EM and hadronic energy measurements up to $|\eta| = 4.9$. The muon spectrometer (MS) comprises separate trigger and high-precision tracking chambers measuring the deflection of muons in a magnetic field generated by the superconducting air-core toroidal magnets. The field integral of the toroids ranges between 2.0 and 6.0 T m across most of the detector. Three layers of precision chambers, each consisting of layers of monitored drift tubes, cover the region $|\eta| < 2.7$, except in the innermost layer of the end-cap region, where layers of small-strip thin-gap chambers and Micromegas chambers both provide precision tracking in the region $1.3 < |\eta| < 2.7$. The muon trigger system covers the range $|\eta| < 2.4$ with resistive-plate chambers in the barrel, thin-gap chambers in the endcap regions, and the aforementioned small-strip thin-gap chambers and Micromegas chambers in the innermost layer of the endcap. The luminosity is measured mainly by the LUCID–2 detector which is located close to the beam pipe. A two-level trigger system is used to select events [29]. The first-level trigger is implemented in hardware and uses a subset of the detector information to accept events at a rate close to 100 kHz. This is followed by a software-based trigger that reduces the accepted rate of complete events to 3 kHz on average, depending on the data-taking conditions. A software suite [30] is used in data simulation, in the

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z -axis along the beam pipe. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upwards. Polar coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the z -axis. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$ and is equal to the rapidity $y = \frac{1}{2} \ln \left(\frac{E+p_z}{E-p_z} \right)$ in the relativistic limit. Angular distance is measured in units of $\Delta R \equiv \sqrt{(\Delta y)^2 + (\Delta \phi)^2}$.

reconstruction and analysis of real and simulated data, in detector operations, and in the trigger and data acquisition systems of the experiment.

3 Signal and background simulation

3.1 Data

The measurements presented in this note are performed using pp collision data recorded with the ATLAS detector between 2022 and 2024, corresponding to an integrated luminosity of 164 fb^{-1} . Data are selected only if all detector components relevant for each analysis channel are in good operating condition. The uncertainty in the integrated luminosity is 1.9% [31], following the methodology described in Ref. [32]. The average number of interactions per bunch crossing (pile-up), averaged over all colliding bunch pairs, is about 42 in 2022, increases to about 51 in 2023, and reaches approximately 57 in 2024.

Events used in the measurement are recorded using a combination of single-lepton, dilepton, and trilepton triggers [33, 34]. The electron triggers used for the 2023 and 2024 data-taking periods exploit the new electromagnetic feature extractors deployed for LHC Run 3 [29]. The transverse momentum thresholds of the single-lepton triggers are 24 GeV for muons and 26 GeV for electrons. The dielectron triggers require a minimum transverse momentum of 17 GeV per electron, while the dimuon triggers require at least 14 GeV per muon. Mixed lepton flavour $e\mu$ triggers combine these thresholds, and trilepton triggers allow for lower thresholds in exchange for a higher lepton multiplicity. For simulated Higgs boson candidates produced via the gluon–gluon fusion process, the resulting total trigger efficiency is 98.1% for signal events passing the final selection described in Section 4.

3.2 Simulated event samples

This analysis uses simulated event samples to model the Higgs boson signal and all relevant background processes, and to derive corrections for detector effects.

3.2.1 Signal modelling

Higgs boson production and decay are simulated for the primary production-modes: gluon–gluon fusion (ggF), vector-boson fusion (VBF, where $V = W, Z$), associated production with a vector boson (VH), top- or bottom-associated production² ($t\bar{t}H$ and $b\bar{b}H$), and associated production with a single top quark (t -channel for tH and tHW)³. Hard-scattering matrix-elements (MEs) for the nominal signal samples are generated with the POWHEG Box v2 [35–39] Monte Carlo (MC) generator. Next-to-leading-order (NLO) accuracy is achieved for the ggF and VH processes with up to one additional parton by employing the MiNLO [40–42] approach. In a subsequent step, the ggF simulation is improved to next-to-next-to-leading-order (NNLO) accuracy for inclusive observables by reweighting the Higgs boson rapidity spectrum in HJ-MiNLO [40–42] to that obtained with HNNLO [43]. The VBF and $t\bar{t}H$ processes are simulated at NLO accuracy in quantum chromodynamics (QCD), while the $b\bar{b}H$ sample is generated in the four-flavour scheme. The loop-induced $gg \rightarrow ZH$ process is generated at leading order in QCD. tH and tHW processes are generated at NLO

² $t\bar{t}H$ and $b\bar{b}H$ are also denoted as ttH and bbH in the figures.

³ The t -channel processes tH and tHW are denoted as tH in the figures.

using the MADGRAPH5_AMC@NLO generator [44]. All signal samples are generated assuming a Higgs boson mass of 125 GeV⁴ and a Higgs boson width of 4.07 MeV.

The signal events are then interfaced to a parton-shower generator for showering, hadronisation, and modelling of the underlying event. All signal samples use the PDF4LHC21 Parton Density Function (PDF) set [45] for the ME calculations. The nominal signal samples are interfaced to PYTHIA 8.309 to simulate the $H \rightarrow ZZ^* \rightarrow 4\ell$ decays and employ the A14 tune for the parton shower [46].

Alternative signal samples are produced to assess uncertainties related to parton-shower and hadronisation modelling. These are obtained by interfacing the same POWHEG events to HERWIG 7 [47] with the H7.1 event tune, using the same PDF setup for the ME calculation. In addition, a simulation of the ggF process is generated using the MADGRAPH5_AMC@NLO generator and showered with PYTHIA 8.312 [48] using the A14 tune. This sample is accurate to NLO in QCD for up to two additional partons merged using the FxFx [44, 49] merging scheme.

The decays of bottom and charm hadrons are simulated using EVTGEN [50].

3.2.2 Signal normalisation and branching ratios

The signal samples are normalised to state-of-the-art theoretical cross-section predictions [51] interpolated to a centre-of-mass energy of 13.6 TeV [52]. The ggF sample is normalised to a N³LO QCD calculation including NLO EW corrections [53–64]. The VBF and VH samples are normalised to an approximate NNLO QCD cross-section with NLO EW corrections [65–75]. The $b\bar{b}H$ contribution is normalised using a calculation combines NLO contributions with the resummation of collinear logarithms up to NNLO [76, 77], and are cross-checked using an alternative matching scheme [78, 79]. The $t\bar{t}H$ cross-section is taken from a calculation accurate to NLO in QCD with NLO EW corrections [80–83].

The Standard Model prediction for the $H \rightarrow ZZ^* \rightarrow 4\ell$ ($\ell = e, \mu$) branching ratio is obtained from PROPHECY4F [84, 85]. This prediction includes the full NLO EW corrections and interference effects, that lead to a branching ratio approximately 10% higher for same-flavour final states (4μ and $4e$) than for different-flavour final states ($2e2\mu$ and $2\mu2e$). The predicted SM branching ratio is $(1.24 \pm 0.03) \times 10^{-4}$ for a Higgs boson mass of 125.09 GeV.

3.2.3 Background modelling

The dominant background contribution arises from continuum $(Z^{(*)}/\gamma^*)(Z^{(*)}/\gamma^*)$ production, and is denoted as ZZ^* . The quark-initiated ($qqZZ^*$) is the dominant process, while the corresponding gluon-initiated process ($ggZZ^*$) provides a smaller contribution of the order of 10%. Background contributions also arise from non-prompt or mis-identified leptons which include the Z production in association with jet activity (Z +jets), top quark pair production ($t\bar{t}$) and WZ diboson production.

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⁴ A shift of 90 MeV is applied to the signal parameterisation used in the fit to the data to match the measured Higgs boson mass of 125.09 ± 0.24 GeV [6].

production in association with jets (Z +jets), top quark pair production ($t\bar{t}$), and WZ diboson production. Additional minor backgrounds include $t\bar{t} + \ell\ell$ and triboson (WWZ , WZZ , ZZZ) production.

Continuum $ZZ^* \rightarrow 4\ell$ production. The nominal $qqZZ^*$ sample is generated using the SHERPA 2.2.14 [86] generator, including off-shell effects. Matrix elements for fully leptonic final states are calculated at NLO accuracy in QCD for up to one additional parton and at LO accuracy for up to three additional partons. These matrix elements are matched and merged with the SHERPA parton shower, based on Catani–Seymour dipole factorisation [87, 88], using the MEPS@NLO prescription [89–92]. Virtual QCD corrections are provided by the OPENLOOPS library [93–95]. The NNPDF3.0_{NNLO} PDF set [96] is used, together with the dedicated tune.

An alternative $qqZZ^*$ sample is generated using the MADGRAPH5_AMC@NLO generator with the NNPDF3.0_{NNLO}, accurate to NLO in QCD for zero and one additional parton merged using the FxFx scheme. In this case, parton showering, hadronisation, and underlying-event modelling are performed with PYTHIA 8.308 using the A14 tune.

The loop-induced $ggZZ^*$ contribution is simulated at NLO accuracy in QCD matched to a parton shower using the POWHEG approach [97] with the NNPDF3.0_{NLO} PDF set. The NLO calculation includes two-loop virtual corrections and loop-squared real-emission contributions, with heavy-quark mass effects included in the virtual corrections. The real-emission matrix elements retain the full top-quark mass dependence [97].

Production of ZZ^* via electroweak vector-boson-scattering-like topologies is simulated at LO using SHERPA 2.2.14 with NNPDF3.0_{NNLO} PDF, with Higgs-mediated contributions removed at generation level.

Since the overall normalisation of the ZZ^* background is derived from collision data, the MC predicted cross-sections are used to only determine the relative fraction of the three different processes.

Top-associated and triboson backgrounds. Minor background contributions include top-associated multilepton production and triboson processes. The production of tWZ , tZq , $t\bar{t}Z$, and $t\bar{t}VV$ is modelled using the MADGRAPH5_AMC@NLO [44] generator at NLO accuracy with the NNPDF3.0_{NLO} PDF set. The events are interfaced to PYTHIA 8.230 using the A14 tune and the NNPDF2.3_{LO} PDF set. The $t\bar{t}W$ process is simulated using SHERPA 2.2.14 with NNPDF3.0_{NNLO} PDF. The combination of these process is referred to as tXX . Triboson processes (WWW , WWZ , WZZ , ZZZ) are simulated using SHERPA 2.2.16, with matrix elements evaluated at NLO accuracy in QCD for up to one additional jet and at LO accuracy for up to three additional jets, merged with the SHERPA parton shower.

Samples for non-prompt lepton background estimates. Simulated samples of Z +jets, $t\bar{t}$, and WZ production are used exclusively to support the data-driven estimation of backgrounds arising from non-prompt or misidentified leptons. The Z +jets samples are generated with SHERPA 2.2.14 following the setup described in Ref [98]. These samples use the NNPDF3.0_{NNLO} PDF set and are normalised to a NNLO QCD prediction [99].

The $t\bar{t}$ process is simulated using the POWHEG BOX v2 generator at NLO accuracy with the NNPDF3.0_{NLO} PDF set and an h_{damp} ⁵ parameter set to $1.5 m_{\text{top}}$ [100]. Events are interfaced to PYTHIA 8.230 using the

⁵ The h_{damp} parameter is a resummation damping factor and one of the parameters that controls the matching of PowHEG matrix elements to the parton shower and thus effectively regulates the high- p_T radiation against which the $t\bar{t}$ system recoils.

A14 tune and the NNPDF2.3_{LO} PDF set. The decays of bottom and charm hadrons are simulated using EVTGEN 2.1.1 [50]. The WZ process is modelled using the SHERPA 2.2.16 generator with the same setup as used for the $qqZZ^*$ sample.

3.2.4 Detector simulation and pile-up modelling

All generated signal and background events are passed through a full simulation of the ATLAS detector based on GEANT4 [101, 102]. Scale factors are applied to the simulation to match the performance of algorithms used to reconstruct physics objects to the one observed in data.

The effects of multiple interactions in the same and neighbouring bunch crossings (pile-up) are modelled by overlaying simulated hard-scattering events with inelastic pp collisions generated using a mixture of EPOS 2.0.1.4 [103] and PYTHIA 8.308 [48]. The EPOS events are generated with the EPOS LHC tune [104], while the PYTHIA events use the A3 tune [105] and the NNPDF2.3_{LO} PDF set. The PYTHIA pile-up events include either a high- p_T jet, a prompt photon, or a lepton from a b -hadron decay, whereas EPOS is used to model the remaining pile-up activity. The individual samples are first reweighted to ensure a smooth transition across jet p_T , and the combined pile-up distribution is subsequently reweighted to match the distribution of the average number of interactions per bunch crossing observed in data.

4 Event selection

Higgs boson candidates are formed by selecting quadruplets of reconstructed and identified charged leptons (electrons and muons). The selection procedure follows that of Ref. [20] and is briefly summarised below.

4.1 Event reconstruction

Event reconstruction begins with the formation of tracks in the ID [106] from measurements in the silicon detectors and the TRT, and in the MS from measurements in the muon chambers. ID tracks are reconstructed for $p_T > 500$ MeV and are used to form vertex candidates along the beam line [107, 108]. Candidate events are required to contain at least one vertex reconstructed from at least two ID tracks. Among all reconstructed vertices in an event, the primary interaction vertex is defined as the one with the largest summed p_T^2 of its associated tracks.

Electron candidates are reconstructed by matching electromagnetic calorimeter clusters to ID tracks [109]. A refit of the ID track, allowing for bremsstrahlung energy losses, is performed to improve the impact-parameter⁶ resolution. Electron identification is based on a likelihood discriminant [110], using a *loose* criteria, that combines track properties, electromagnetic shower-shape variables, and the quality of the track-cluster matching. Electrons are required to satisfy $E_T > 7$ GeV and $|\eta| < 2.47$, where η is calculated using calorimeter clusters. The electron energy is calibrated following the procedure described in Ref. [109]. Photons are reconstructed from EM calorimeter signals exceeding noise thresholds using dynamic topological clustering, with additional clusters included for photon conversion products when

⁶ The transverse impact parameter d_0 of a charged-particle track is defined in the transverse plane as the distance of closest approach to the beam spot. The longitudinal impact parameter z_0 is defined as the distance along the beam axis between this point and the primary vertex.

applicable [109]. The photon candidates are calibrated [111] and identified by an algorithm that relies on the shape of the EM showers reconstructed in the calorimeter [110].

Muon candidates are reconstructed in the pseudorapidity range $|\eta| < 2.5$ by matching tracks reconstructed in the MS to ID tracks. In the region $|\eta| < 0.1$, where the MS is only partially instrumented, muons are additionally reconstructed from ID tracks matched either to partial MS track segments or to calorimeter energy deposits consistent with a minimally ionising particle (calorimeter-tagged muons) [112]. Muons are required to have $p_T > 5$ GeV, except for calorimeter-tagged muons, which must satisfy $p_T > 15$ GeV. A set of *Loose* identification criteria is applied to suppress muons from hadron decays and misidentified tracks [112]. The muon momentum is calibrated using the procedure described in Ref. [113].

Electron and muon candidates are associated with the primary vertex using their impact parameters. Muon candidates with $|d_0| > 1$ mm are rejected to suppress cosmic-ray background. Both electron and muon candidates are required to satisfy $|z_0 \sin \theta| < 0.5$ mm.

Jet candidates are reconstructed using the particle-flow (PFlow) algorithm [114], which combines calorimeter energy deposits and inner-detector tracks into particle-flow objects that are clustered into jets using the anti- k_t [115] jet algorithm with radius parameter $R = 0.4$. Jets are calibrated using simulation with corrections obtained by using *in situ* techniques in data [116]. After calibration, jets are required to satisfy $p_T > 30$ GeV and $|\eta| < 4.5$. To suppress jets originating from pile-up interactions, a neural-network-based algorithm [117, 118] is used to associate jets with $p_T < 60$ GeV to the primary vertex.

Jets containing b -hadrons are identified (b -tagged) using the GN2 transformer-based neural network algorithm [25]. Compared with previous algorithms, GN2 does not rely on lower-level flavour-tagging algorithms and uses a single training to optimise all input features. It demonstrates improved background rejection at a given efficiency compared with the previous tagger [25]. Two b -tagging working points are used in this analysis, one with an b -tagging efficiency of approximately 85% and a tighter working point with an efficiency of about 65%.

Ambiguities are resolved if electron, muon, or jet candidates overlap in geometry or share the same detector information. If the two calorimeter energy clusters from the two electron candidates overlap, the electron with the higher E_T is retained. If a reconstructed electron and muon share the same ID track, the muon is rejected if it is calorimeter-tagged; otherwise the electron is rejected. Reconstructed jets geometrically overlapping in a cone of radial size $\Delta R = 0.1$ (0.2) with a muon (an electron) are also removed.

The missing transverse momentum vector, $\vec{p}_T^{\text{miss}}$, is defined as negative vector sum of the transverse momenta of all the identified and calibrated leptons and jets and the remaining unclustered energy, where the latter is estimated from low- p_T tracks associated with the primary vertex but not assigned to any lepton, photon, hadronically decaying τ -lepton or jet candidate [119]. The missing transverse momentum (E_T^{miss}) is defined as the magnitude of $\vec{p}_T^{\text{miss}}$.

4.2 Higgs boson candidate formation

Higgs boson candidates are formed from pairs of same-flavour, opposite-charge (SFOC) electron or muon pairs. Each such combination, referred to as a quadruplet, is required to include the leptons that triggered the event, as described in Section 3.1. The three highest- p_T leptons in the quadruplet are required to satisfy $p_T > 20, 15, \text{ and } 10$ GeV, respectively.

Further requirements are imposed on the invariant masses of the SFOC lepton pairs. For each quadruplet, the lepton pair with invariant mass m_{12} closest to the Z boson mass [120] is defined as the leading pair, while the remaining SFOC pair, with invariant mass m_{34} second-closest to the Z boson mass, is defined as the subleading pair. On the selected SFOC lepton pairs, additional selection criteria are applied on the angular separation between leptons, a J/ψ veto, the lepton transverse impact-parameter significance ($|d_0|/\sigma(d_0)$), and the quality of the four-lepton vertex. The four leptons are required to be fitted to a common vertex and to be compatible with originating from the same interaction point. In addition, track- and calorimeter-based isolation requirements are applied on the lepton candidates in the pairs to suppress the non-prompt contributions. Those requirements are described in detail in [20]. The full set of requirements is summarised in Table 1.

If more than one quadruplet in an event satisfies all selection criteria, the candidates containing the leading lepton pair closest to the Z boson mass are selected. If more than one quadruplet with the same leading lepton pair but different subleading pairs pass this criteria, the one corresponding to the lepton-flavour combination with the highest total selection efficiency is chosen, which is about 28%, 23%, 14%, and 12% for the 4μ , $2e2\mu$, $2\mu2e$, and $4e$, final states, respectively. $2e2\mu$ and $2\mu2e$ final states are distinguished by the flavour of the leading lepton pair.

If after this selection, an additional lepton, not used in the quadruplet, with $p_T > 12$ GeV is present in the event, it may originate from leptonic decays of VH and $t\bar{t}H$ production-modes. To improve the mass resolution in this scenario, a decay-side ME discriminant is calculated at leading order using MADGRAPH5 for each quadruplet in the event, with the reconstructed lepton four-momenta used as inputs [8], and the quadruplet with the highest ME value is selected.

The selected Higgs boson candidates are further refined by accounting for final-state radiation (FSR) photons, using the procedure described in Refs. [8, 9]. For electrons, the reconstruction already includes the recovery of nearby FSR photons within $\Delta R < 0.15$ [121]; therefore, only the recovery of far FSR photons is applied to electron candidates in this analysis. For muons, both nearby and far FSR photons are explicitly recovered.

5 Background estimation

5.1 Background process with prompt leptons

The dominant background contribution arises from continuum ZZ^* production initiated by quarks or gluons. Unlike the Higgs signal, the $m_{4\ell}$ spectrum of the ZZ^* background does not exhibit a resonant structure near the Higgs mass, m_H . This feature is exploited to constrain the normalisation of the ZZ^* background directly from data.

The shapes of the $m_{4\ell}$ templates and the differential distributions for the ZZ^* background are taken from the simulation described in Section 3. The relative contributions of the individual ZZ^* subprocesses are fixed to their simulated expectations, while the overall normalisation is left free in the statistical model through one or more scale factors, N/N_{MC} , defined with respect to the MC-predicted yields. The binning of these scale factors differs between the two measurements and is described below.

The ZZ^* normalisation is constrained using signal-depleted sideband regions defined by $105 < m_{4\ell} < 115$ GeV and $130 < m_{4\ell} < 160$ GeV. These regions are representative of the continuum background

Table 1: Summary of the event selection requirements for leptons and Higgs boson candidates. SFOC lepton pairs are same-flavour opposite-charge lepton pairs. For the subleading lepton-pair mass requirement, $m_{\text{threshold}}$ is 12 GeV for $m_{4\ell} < 140$ GeV, and increases linearly to 50 GeV at $m_{4\ell} = 190$ GeV, remaining constant after [8].

Leptons	
Muons	$p_T > 5$ GeV, $ \eta < 2.5$
Electrons	$E_T > 7$ GeV, $ \eta < 2.47$
Jets	$p_T > 30$ GeV, $ \eta < 4.5$
Lepton selection and pairing	
Lepton kinematics	$p_T > 20, 15, 10$ GeV
Leading pair (m_{12})	SFOC lepton pair with smallest $ m_Z - m_{\ell\ell} $
Subleading pair (m_{34})	remaining SFOC lepton pair with smallest $ m_Z - m_{\ell\ell} $
Event selection (at most one Higgs boson candidate per channel)	
Mass requirements	$50 \text{ GeV} < m_{12} < 106 \text{ GeV}$ and $m_{\text{threshold}} < m_{34} < 115 \text{ GeV}$
Lepton separation	$\Delta R(\ell_i, \ell_j) > 0.1$
J/ψ veto	$m(\ell_i, \ell_j) > 5$ GeV for all SFOC lepton pairs
Impact parameter	$ d_0 /\sigma(d_0) < 5$ (3) for electrons (muons)
Isolation	Track- and calorimeter-based isolation for all four leptons
Vertex selection	$\chi^2/N_{\text{dof}} < 6$ (9) for 4μ (other channels)
If extra lepton with $p_T > 12$ GeV	quadruplet with largest matrix-element value

process under the Higgs boson peak and avoids the region of the $ZZ^* \rightarrow 4\ell$ before 105 GeV and the fully resonant ZZ production after the 160 GeV.

For the fiducial cross-section measurements, including differential observables, the fit is performed over the full mass range $105 < m_{4\ell} < 160$ GeV. For differential measurements, the ZZ^* normalisation is determined independently in each bin of the measured observable. To reduce the impact of statistical fluctuations in categories with low expected statistics, neighbouring bins are combined when extracting the corresponding normalisation parameters. For the production-mode cross-section measurements in the STXS framework, the same sideband regions are combined in $m_{4\ell}$ but are further categorised according to the reconstructed jet multiplicity ($N_{\text{jets}} = 0, 1$, and ≥ 2) to reduce theoretical assumption on extrapolating results between different regions. Signal-region categories with the same jet multiplicity are fitted simultaneously with their corresponding sideband regions, using a common ZZ^* normalisation parameter for each jet-multiplicity category.

This strategy substantially reduces theoretical uncertainties in the ZZ^* background prediction for both measurements, as the statistical model primarily relies on data to constrain the normalisations, leaving only the relative normalisation between the signal and sideband regions and the modelling of the differential shapes dependent on theoretical assumptions.

Top-associated background processes are treated differently in the two measurement strategies. In the fiducial and differential cross-section measurements, their contribution is small and is therefore taken directly from simulation. In the production-mode cross-section measurements analysis, where sensitivity to $t\bar{t}H$ production is enhanced, a dedicated control region is included in the fit to constrain the normalisation of these processes. A single normalisation parameter, N^{tXX}/N_{MC} , is determined in a simultaneous fit to

the $t\bar{t}H$ -like signal regions and the control region, defined following the Run 2 analysis strategy described in Ref. [8]. In this control region, the upper bound of the $m_{4\ell}$ selection is extended to 350 GeV to improve the statistical precision of the constraint, while avoiding the fully resonant production.

Contributions from triboson production are taken directly from simulation in all regions considered.

5.2 Background process with non-prompt leptons

A small contribution to the selected 4ℓ sample arises from events with two prompt leptons accompanied by additional non-prompt leptons from heavy-flavour hadron decays, light-jet misidentification, or photon conversions (including final-state radiation), primarily from Z +jets, $t\bar{t}$, and WZ production. Non-prompt muon typically originate more predominantly from heavy-flavour hadron decays, while non-prompt electrons originate from light-jet misidentification and photon conversions. Such non-prompt leptons are typically produced around or nearby hadronic activity and therefore have a higher probability to fail the isolation and/or impact-parameter requirements used in the signal selection. This background is referred to as *reducible* and is estimated with a data-driven fake-factor method, following approaches used in other four-lepton final-state analyses [122, 123].

The method is based on two nested lepton definitions: *baseline* leptons satisfy the full signal selection described in Section 4, except for the isolation and impact-parameter requirements, while *tight* leptons additionally satisfy these requirements. The fake efficiency f is defined as the probability for a non-prompt (fake) baseline lepton to also satisfy the tight requirements, and is measured in data after subtracting the expected prompt-lepton contamination. The fake factor (F) is then defined as

$$F = \frac{f}{1 - f}, \quad (1)$$

relating the yield of events with tight fake leptons to the yield of events in which the corresponding leptons fail the tight requirements.

The fake factors are measured in two control regions enriched in reducible background: a Z +jets-dominated region (CR_Z) and a $t\bar{t}$ -dominated region (CR_{top}). In both regions, a tight lepton pair is required together with an additional baseline lepton. In CR_Z , the tight pair is required to be same-flavour opposite-sign with an invariant mass close to m_Z , while in CR_{top} the tight pair is required to be an $e\mu$ pair and the event must contain at least one b -tagged jet using the working point with 85% efficiency, implying that the 0-jet category is not available in CR_{top} and is measured using CR_Z only. Prompt-lepton contamination in the measurement regions is dominated by WZ production. Its contribution is estimated in each CR applying a scale factor (SF) to correct the WZ MC estimation with data. This SF is extracted from a dedicated region (CR_{WZ}), that requires a SFOS tight lepton pair with an invariant mass close to m_Z and one extra tight lepton.

The fake factors are parameterised separately for electrons and muons in a three-dimensional map of the lepton p_T , $|\eta|$, and jet multiplicity. The final fake efficiency in the signal region (SR) is obtained by combining the measurements from CR_Z and CR_{top} with equal weights,

$$f^{\text{SR}} = 0.5 f^{\text{CR}_Z} + 0.5 f^{\text{CR}_{\text{top}}}, \quad (2)$$

while for the 0-jet category $f_{0j}^{\text{SR}} = f_{0j}^{\text{CR}_Z}$. While the compositions for the non-prompt lepton in the signal region are more similar to those observed in CR_{top} , an equal-weighted average is used to ensure a robust

estimate as weighting according to the composition increases sensitivity to statistical variation from CR_{top} , which has significantly lower event yields than CR_Z . However, as described in Section 7, the full difference between two regions is included as a conservative systematic uncertainty to account for the composition difference. This systematic uncertainty has a negligible impact on the final results.

The reducible-background yield in the signal region is estimated by applying fake-factor event weights to data events that pass the nominal quadruplet selection but contain at least one baseline lepton failing the tight requirements, after subtracting residual prompt contributions from simulation.

The procedure is validated in two additional regions that are kinematically close to the signal region but enriched in the reducible background: a same-charge region requiring the subleading lepton pair to have the same charge, and an $e\mu$ region requiring one lepton pair to be an $e\mu$ pair, with all other selections identical to the signal region. To reduce statistical fluctuations, a smoothing procedure is applied and the corresponding statistical uncertainty is evaluated using pseudo-datasets.

6 Analysis overview

The analysis performs two complementary measurements of the Higgs boson cross-section in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel. In the first approach, inclusive and differential cross-sections are measured in a fiducial phase space designed to closely mirror the detector-level selection, thereby minimising extrapolation and reducing the dependence on theoretical assumptions. In this measurement, the relative contributions of the different Higgs boson production-modes are fixed to the Standard Model expectation when deriving acceptance and migration corrections. Signal yields are extracted from maximum-likelihood fits to the reconstructed $m_{4\ell}$ spectrum, and detector effects are corrected using a likelihood-based matrix unfolding procedure [9] to obtain particle-level results that are as model independent as possible and sensitive to deviations from the Standard Model.

The second complementary approach aims to measure the underlying Higgs boson production mechanisms. Production-mode cross-sections are measured using event categories and multivariate discriminants optimised for separation of ggF, VBF, VH , and $t\bar{t}H$ contributions, and are interpreted within the STXS framework. The STXS approach provides a common set of phase-space regions that enables increasingly granular measurements, enhancing sensitivity to differences in production kinematics and potential BSM effects.

The following sections describe these two strategies in detail, first presenting the fiducial inclusive and differential cross-section measurement and then the production-mode measurement in the STXS framework. Although both approaches provide an inclusive Higgs boson production cross-section, they do so using complementary combinations and are impacted by different model dependences. In the fiducial analysis, the inclusive cross-section is obtained by combining measurements performed separately in the four decay final states, and depends on the modelling of the relative effects between the 4μ , $2e2\mu$, $2\mu2e$, and $4e$ channels. In the STXS measurement, the inclusive cross-section is obtained by summing the fitted production-mode contributions across the reconstructed categories, and is sensitive to the modelling of the relative signal composition across jet-multiplicity regions and categories.

6.1 Inclusive and differential cross-sections

6.1.1 Fiducial phase space and observables

The fiducial cross-sections are defined at particle level in a phase space chosen to closely match the detector-level event selection, thereby reducing the size of model-dependent acceptance extrapolations and minimising theoretical and experimental uncertainties associated with the unfolding procedure [9]. The fiducial phase space definition is outlined in Table 2. Leptons are defined as final-state electrons and muons that do not originate from hadron or τ -lepton decays, after FSR [124]. Leptons are “dressed” by adding the four-momenta of photons within a cone of $\Delta R = 0.1$ around the lepton direction (excluding photons from hadron decays) to the lepton four-momentum [9] to recover energy lost to FSR. Dressed leptons are required to satisfy $|\eta| < 2.7$ and $p_T > 5$ GeV irrespective of flavour to simplify comparisons with theoretical predictions. Particle-level jets are reconstructed from final state neutral and charged particles using the anti- k_t algorithm with radius parameter $R = 0.4$ and are required to satisfy $p_T > 30$ GeV and $|y| < 4.4$. Overlapping jets are removed if $\Delta R(\ell, \text{jet}) < 0.1$ for with any selected lepton [9]. Higgs boson candidates are formed from SFOC lepton pairs using the quadruplet selection described in Section 4.2, with the exception of the isolation, impact-parameter, and vertex requirements. Events enter the fiducial phase space if a candidate satisfying the mass and kinematic requirements is reconstructed, including $105 < m_{4\ell} < 160$ GeV [9].

Table 2: List of event selection requirements at particle level which define the fiducial phase space for the cross-section measurement. SFOC lepton pairs are same-flavour opposite-charge lepton pairs.

Leptons and jets	
Leptons	$p_T > 5$ GeV, $ \eta < 2.7$
Jets	$p_T > 30$ GeV, $ y < 4.4$
Lepton selection and pairing	
Lepton kinematics	$p_T > 20, 15, 10$ GeV
Leading pair (m_{12})	SFOC lepton pair with smallest $ m_Z - m_{\ell\ell} $
Subleading pair (m_{34})	remaining SFOC lepton pair with smallest $ m_Z - m_{\ell\ell} $
Event selection (at most one quadruplet per event)	
Mass requirements	$50 \text{ GeV} < m_{12} < 106 \text{ GeV}$ and $12 \text{ GeV} < m_{34} < 115 \text{ GeV}$
Lepton separation	$\Delta R(\ell_i, \ell_j) > 0.1$
Lepton/Jet separation	$\Delta R(\ell_i, \text{jet}) > 0.1$
J/ψ veto	$m(\ell_i, \ell_j) > 5$ GeV for all SFOC lepton pairs
If extra lepton with $p_T > 12$ GeV	Quadruplet with largest matrix element value
Mass window	$105 \text{ GeV} < m_{4\ell} < 160 \text{ GeV}$

Fiducial production cross-sections are measured inclusively and separately for each $H \rightarrow ZZ^* \rightarrow 4\ell$ final state ($4\mu, 2e2\mu, 2\mu2e, 4e$). Differential fiducial cross-sections are measured for eight observables sensitive to the production and decay kinematics of the Higgs boson. The transverse momentum of the four-lepton system, $p_T^{4\ell}$, provides a stringent test of perturbative QCD calculations and is sensitive to modifications of Higgs boson production dynamics. The rapidity of the four-lepton system, $|y_{4\ell}|$, probes the parton distribution functions of the colliding protons and is also affected by QCD radiative corrections. The invariant masses of the leading and subleading SFOC lepton pairs, m_{12} and m_{34} , are sensitive to higher-order electroweak effects and to the decay kinematics of $H \rightarrow ZZ^* \rightarrow 4\ell$. The decay angle between

the two Z bosons ϕ , is sensitive to the Higgs boson spin and parity and to BSM contributions in the HZZ vertex. Jet activity is characterised by the jet multiplicity, N_{jets} , which is sensitive to QCD radiation and to the relative contributions of different production mechanisms. For events with at least two jets, the azimuthal separation of the two leading jets, $|\Delta\phi_{jj}|$, provides additional sensitivity to production dynamics through the structure of associated QCD radiation. In addition, a double-differential measurement of $p_{\text{T}}^{4\ell}$ versus m_{34} is performed, which enhances sensitivity to BSM tensor structure couplings effects affecting both the production and decay of the Higgs boson.

6.1.2 Signal extraction and unfolding

The fiducial cross-section σ_j^{fid} in each bin j of a differential distribution (or, for the inclusive measurement, in each decay final state) is obtained from a binned maximum-likelihood fit of signal and background templates to the $m_{4\ell}$ spectrum. The fit is performed separately in each reconstruction bin i of the corresponding observable. The expected number of events in reconstruction bin i , expressed as a function of $m_{4\ell}$, is given by

$$N_i(m_{4\ell}) = \sum_j \sigma_j^{\text{fid}} \cdot \mathcal{P}_i(m_{4\ell}) \cdot r_{ij} \cdot (1 + f_i^{\text{nonfid}}) \cdot \mathcal{L} + N_i^{\text{bkg}}(m_{4\ell}) \quad (3)$$

where $\sigma_j^{\text{fid}} = \sigma_j \cdot a_j \cdot \mathcal{B}$. Here, σ_j is the total production cross-section in particle-level bin j , a_j is the particle-level acceptance of the fiducial selection in that bin, and $\mathcal{B}$ is the branching ratio. The integrated luminosity is denoted by $\mathcal{L}$, and $N_i^{\text{bkg}}(m_{4\ell})$ is the background contribution in reconstruction bin i . The term $\mathcal{P}_i(m_{4\ell})$ is the signal $m_{4\ell}$ template for reconstruction bin i taken from simulation. The response matrix element r_{ij} corresponds to the probability that an event generated in fiducial bin j is reconstructed in bin i . The fiducial leakage factor, f_i^{nonfid} , accounts for the fraction of events produced outside the fiducial phase space that are nevertheless reconstructed within the detector-level signal region. The factors a_j , r_{ij} , and f_i^{nonfid} are derived from simulated signal samples, assuming Standard Model production-mode fractions within their associated theoretical uncertainties.

As outlined in Section 5, the background contribution $N_i^{\text{bkg}}(m_{4\ell})$ includes the continuum ZZ^* component and smaller prompt and non-prompt backgrounds. To minimise theory dependence, the overall ZZ^* yield is constrained directly from data by including $m_{4\ell}$ sideband regions in the likelihood fit. The remaining prompt and non-prompt contributions are taken from simulation and the data-driven method respectively.

The binning choice mostly follows the previous measurement [9], which was designed to ensure an expected statistical significance per bin of more than two standard deviations for the SM signal process and to minimise migrations between bins. These same criteria are used to define the binning of the new observable $p_{\text{T}}^{4\ell}$ versus m_{34} . In most cases, the bin width is more than twice the experimental resolution. The response matrices for all the variables considered have been checked to be well-conditioned, therefore no regularisation of the unfolding was applied.

The response matrix for p_{T}^H and N_{jets} are shown in Figure 1. The bin-to-bin migrations are quantified by the *purity*, defined as the fraction of reconstructed events in a given bin that originate from the corresponding particle-level bin. The purity is mostly larger than 80% for the observables related to the Higgs decay kinematics, the $p_{\text{T}}^{4\ell}$, $y_{4\ell}$, m_{12} , m_{34} , ϕ and consequently $p_{\text{T}}^{4\ell}$ versus m_{34} . Instead, for the jet-related observables, N_{jets} and $|\Delta\phi_{jj}|$, migrations are larger due to the jet energy resolution and residual pile-up effects, reducing for example the purity in the highest jet-multiplicity bin to 66%. The fiducial leakage mostly ranges from 1% to 2%, depending on the bin of the unfolded observable or the decay final state. Only the first bin of m_{12} and the last bin of m_{34} observables reach values of 4% and 6% respectively.

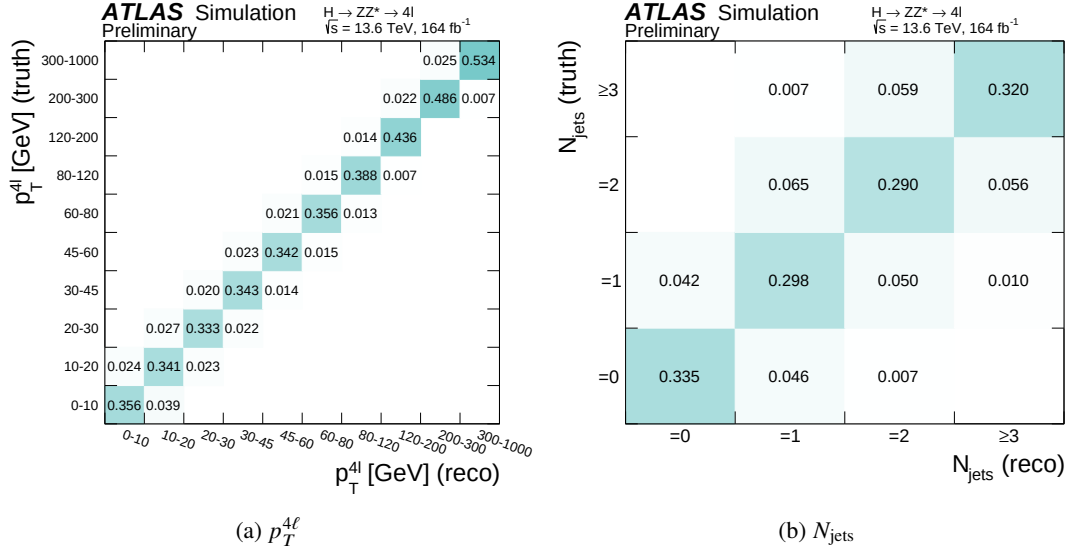


Figure 1: Response matrices, derived from simulation, for (a) the transverse momentum of the four-lepton system $p_T^{4\ell}$ and (b) the jet multiplicity N_{jets} . The response matrix is used in the unfolding procedure (Eq. 3). The elements r_{ij} correspond to the probability that an event generated in fiducial bin j is reconstructed in bin i . Only reconstructed events that are matched to particle-level (“truth”) events are included. Bins with values below 0.005 are omitted for clarity.

6.2 Production-mode cross-sections

The cross-sections are also probed separately for four different Higgs boson production-modes: gluon–gluon fusion (ggF), vector-boson fusion (VBF) and associated production with vector bosons (VH , where $V = W$ or Z) or top quark pairs ($t\bar{t}H$).

6.2.1 Simplified template cross-sections

In the STXS framework [22, 23, 125], Higgs boson production cross-sections are measured in exclusive particle-level regions (production bins) designed to minimise the impact of theoretical uncertainties that would otherwise enter through acceptance extrapolations, while retaining sensitivity to production dynamics and potential BSM effects. Two alternative STXS binnings are considered: a production-mode (also referred to as Stage-0) measurement, in which cross-sections are extracted for the four dominant production-modes (ggF, VBF, VH , and $t\bar{t}H$), and a Reduced Stage-1.2 measurement that provides additional kinematic information by subdividing production according to jet activity and Higgs boson kinematics. Bins are merged with respect to the full binning where required by the statistical precision. All production bins are defined for Higgs bosons with rapidity $|y_H| < 2.5$, and no requirements are placed on particle-level leptons. Particle-level jets are used for this categorisation and are built using the anti- k_t algorithm with $R = 0.4$ from stable particles (lifetime $c\tau > 10$ mm) and are required to satisfy $p_T > 30$ GeV with no rapidity restriction; Higgs decay products and leptons/neutrinos from leptonic V decays are excluded from the jet clustering, while hadronic V decay products are included [23].

In the production-mode measurement, contributions from $b\bar{b}H$ are included in the ggF bin and contributions from $t\bar{t}H$ are included in the $t\bar{t}H$ bin, with their relative fractions fixed to SM predictions. VH events with

hadronic V decays are retained in a dedicated VH bin [22]. For the Reduced Stage-1.2 measurement, the ggF and $gg \rightarrow ZH$ (hadronic Z) contributions are split by p_T^H and jet multiplicity, including a dedicated high- p_T^H bin with $p_T^H \geq 200$ GeV to enhance sensitivity to BSM effects. The electroweak qqH contribution (including VBF and hadronic VH) is further partitioned using p_T^H and the dijet invariant mass m_{jj} , defining a BSM-sensitive region with $p_T^H \geq 200$ GeV and $m_{jj} > 350$ GeV, a VH -like region with $60 < m_{jj} < 120$ GeV, and a remaining region targeting VBF, while VH with leptonic V decays is treated separately as VH -Lep. The $t\bar{t}H$ production bin definition is unchanged with respect to the production-mode interpretation.

The exact definitions of the STXS production bins and the corresponding labels used in this analysis are shown on the left grayed panel of Figure 2.

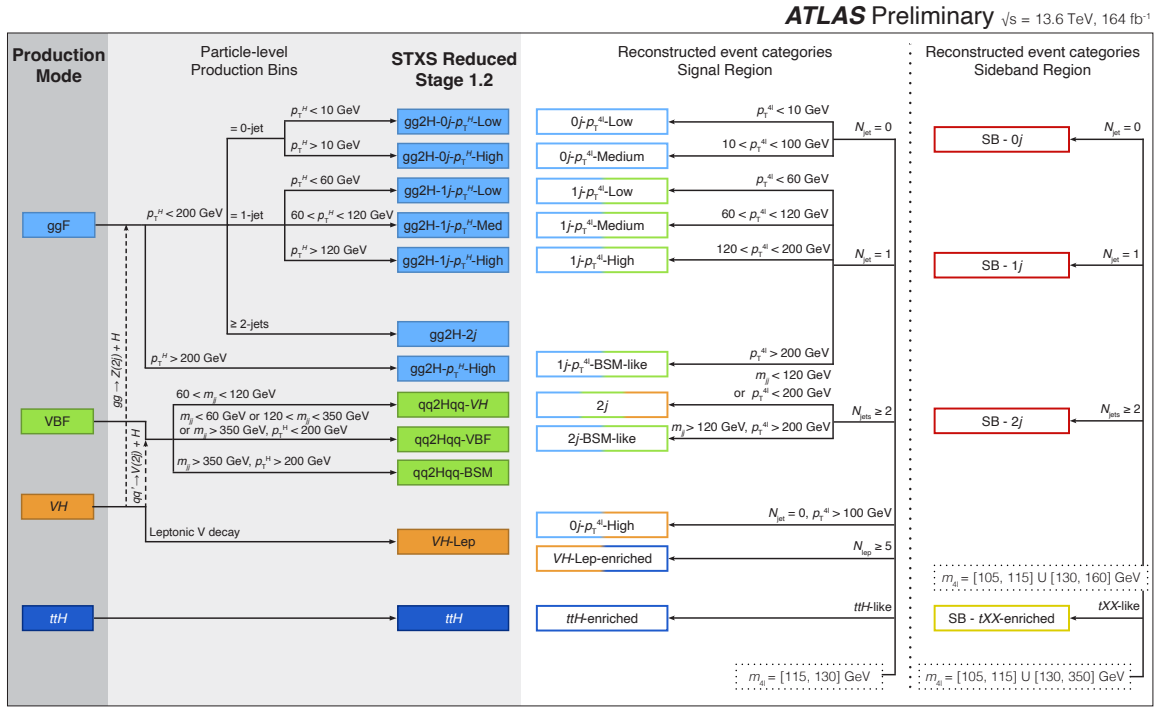


Figure 2: Two sets of exclusive phase-space regions (production bins) are defined at particle level for the measurement of the Higgs boson production cross-sections: the production-mode Stage 0 and the Reduced Stage 1.2 scheme (left and middle-left shaded panels). The corresponding reconstructed event categories are shown for signal (middle-right panel) and sidebands (right panel). The description of the production bins is given in Section 6.2.1, while the reconstructed signal-region and sideband event categories are described in Section 6.2.2. The $b\bar{b}H$ ($t\bar{t}H$) contribution is included in the ggF ($t\bar{t}H$) production bins. The colours of each reconstructed event-category box indicate the approximate contributions from the relevant production processes. The reconstructed signal categories are defined sequentially from bottom to top, such that each event is assigned to the first category whose selection criteria are satisfied.

6.2.2 Event categorisation

Signal-region events with a reconstructed Higgs boson candidate in $115 < m_{4\ell} < 130$ GeV are assigned to mutually exclusive categories sensitive to different production bins in the STXS framework, in a sequential procedure, following the ordering shown in Figure 2. Events are first tested against the $t\bar{t}H$ -enriched selection, which targets $t\bar{t}H$ production using requirements on additional leptons and/or high jet and b -tag multiplicities. In particular, events are selected if they contain at least one additional isolated lepton with $p_T > 12$ GeV and satisfy at least one of the following: at least two b -tagged jets at the 85% working point, at least five jets with at least one b -tagged jet at the 85% working point, or at least two jets with at least one b -tagged jet at the 65% working point. Events without an additional isolated lepton can also enter the $t\bar{t}H$ -enriched selection if they satisfy tighter jet and b -tag requirements, namely at least five jets with at least two b -tagged jets at the 85% working point, or at least four jets with at least one b -tagged jet at the 65% working point.

Events with additional isolated leptons that do not satisfy the $t\bar{t}H$ -enriched selection define the VH category targeting leptonic vector-boson decays (VH -Lep-enriched). The remaining events are then categorised by jet multiplicity ($N_{\text{jets}} = 0, 1, \text{ or } \geq 2$) and further subdivided using p_T^H and, for multi-jet events, the dijet invariant mass m_{jj} to target specific STXS production bins. Events with at least two jets are separated into a BSM-like category at high p_T^H and large m_{jj} and a complementary category that contains the bulk of the VBF/ VH contributions in addition to ggF. Both the 0-jet and 1-jet regions are further split into low-, medium-, and high- p_T^H intervals.

Sideband categories in $m_{4\ell}$ are defined to constrain the dominant non-resonant ZZ^* background in different jet multiplicity regions (SB-0j, SB-1j, and SB-2j) and the top-associated tXX background (SB- tXX -enriched) as described in Section 5.

The definitions of the reconstruction categories and the corresponding names used in this analysis are shown on the right panel of Figure 2. The composition of production-modes per reconstructed event category of these categories is shown in Figure 3.

6.2.3 Multivariate discriminants and Signal Extraction

In the STXS measurement, several Higgs boson production mechanisms contribute to each reconstructed event category, together with multiple background processes. To enhance the separation power beyond the cut-based categorisation, multivariate discriminants are constructed using neural networks (NNs) based on a Set Transformer architecture [126]. The networks are trained independently in each reconstructed category and their binned outputs are used as the fit observables. The NN inputs are organised into event-level global observables and sets of reconstructed physics objects. Lepton and jet information is provided as variable-length sets to Set Transformer blocks, which provide permutation-invariant processing through attention mechanisms [24], while global variables are combined with the learned set representations in a classifier head. The specific feature choices and the corresponding training targets follow the strategy outlined in Table 3. Good agreement between data and simulation is observed for the input variables used in the NN training, as illustrated in Figure 10. In addition, validation studies are performed in ZZ^* -enriched control regions to verify that the NN outputs are well modelled and that no overtraining or simulation-specific features are learned. In categories with limited statistical power, no NN discriminant is applied and only the observed yield enters the fit.

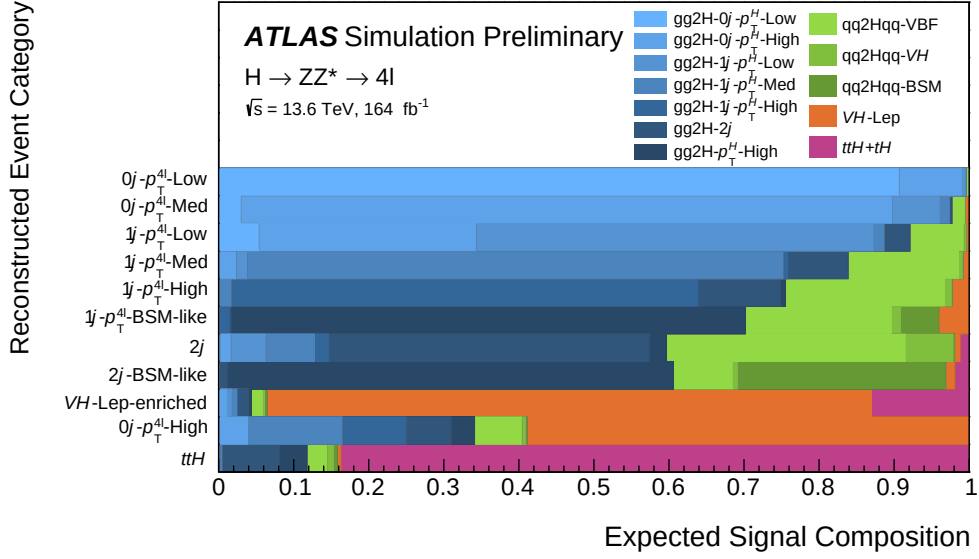


Figure 3: Standard Model signal composition in terms of the Reduced Stage-1.2 production bins in each reconstructed event category. The $b\bar{b}H$ and tH contributions are included in the ggF and $t\bar{t}H$ production bins, respectively.

The STXS cross-sections are obtained from a simultaneous binned likelihood fit to the discriminant distributions across all reconstructed signal categories defined in Section 6.2.2, with the dominant background processes constrained as described in Section 5. The parameters of interest are collected in a vector $\vec{\sigma}$ and are related to the reconstructed event yields N_{ik} in a histogram bin k of the discriminating observable in reconstructed category i via

$$N_{ik} = \vec{A}_{ik} \cdot \vec{\sigma} \cdot \mathcal{B} \cdot \mathcal{L} + N_{ik}^{\text{bkg}}, \quad (4)$$

where $\mathcal{B}$ is the $H \rightarrow ZZ^* \rightarrow 4\ell$ branching ratio, $\mathcal{L}$ is the integrated luminosity, and N_{ik}^{bkg} is the expected background contribution. The vector $\vec{A}_{ik}$ contains the signal selection factors that map particle-level STXS bins to reconstructed categories and discriminant bins, and is determined from MC simulation. In the Stage-0 interpretation, the components of $\vec{\sigma}$ (and thus of $\vec{A}_{ik}$) correspond to the chosen set of production-mode cross-sections, while in the Reduced Stage-1.2 interpretation they correspond to bins defined in Section 6.2.1. In both cases, each factor A_{ik}^j is defined as the fraction of generated signal events in STXS bin j that satisfy the reconstruction and event selection and populate histogram bin k of the discriminant observable in reconstructed category i .

6.3 Statistical method

All measurements are performed with a binned profile-likelihood-ratio fit [129]. The likelihood is built as product of Poisson terms for the fitted discriminant distributions in each differential bin or STXS category, and includes nuisance parameters that account for experimental and theoretical systematic uncertainties affecting the normalisation and shapes of the signal and background templates (Section 7). The parameters of interest are the cross-sections and the background normalisation factor introduced in Sections 6.1 and 6.2 and are extracted by minimising $-2 \ln \Lambda$, where Λ is the profile likelihood ratio. The asymptotic

Table 3: Input variables used for the neural-network trainings in the reconstructed STXS categories that employ binned NN discriminants. The inputs are grouped into event-level *global* variables and object-level variables provided as variable-length *lepton* and *jet* sets to Set Transformer blocks. Here, $p_T^{4\ell}$ and $\eta_{4\ell}$ are the transverse momentum and pseudo-rapidity of the four-lepton system; m_{12} and m_{34} are the invariant masses of the leading and subleading same-flavour opposite-charge lepton pairs; D_{ZZ^*} denotes the kinematic discriminant against non-resonant ZZ^* production; $\cos \theta^*$, $\cos \theta_1$, and ϕ_{ZZ} are decay-angle observables defined as in Ref. [127]; p_T^j and η_j are the transverse momentum and pseudorapidity of the leading jet; m_{jj} is the dijet invariant mass; $p_T^{4\ell jj}$ is the transverse momentum of the four-lepton plus dijet system; and $\eta_{ZZ}^{\text{zepp}} = |\eta_{4\ell} - \frac{\eta_{j1} + \eta_{j2}}{2}|$ is the Zeppenfeld variable of the four-lepton system with respect to the two leading jets [128]. $\Delta R_{4\ell, j}$ is the angular separation between the four-lepton system with respect to the leading jet, while E_T^{miss} is the measured missing transverse energy.

Category	Process	Global inputs	Lepton set inputs	Jet set inputs	Discriminant
0j- $p_T^{4\ell}$ -Low 0j- $p_T^{4\ell}$ -Med	ggF, ZZ^*	$\eta_{4\ell}, p_T^{4\ell}, D_{ZZ^*}, m_{12}$ $m_{34}, \cos \theta^*, \cos \theta_1, \phi_{ZZ}$	p_T^ℓ, η_ℓ	–	NN_{ggF}
1j- $p_T^{4\ell}$ -Low 1j- $p_T^{4\ell}$ -Med	ggF, VBF, ZZ^*	$p_T^{4\ell}, p_T^j, \eta_j, D_{ZZ^*}$ $\Delta R_{4\ell, j}, \eta_{4\ell}, E_T^{\text{miss}}$	p_T^ℓ, η_ℓ	–	$\text{NN}_{\text{ggF+VBF}}$ for $\text{NN}_{ZZ} < 0.2$ NN_{ZZ} for $\text{NN}_{ZZ} > 0.2$
1j- $p_T^{4\ell}$ -Med 1j- $p_T^{4\ell}$ -High	ggF, VBF, ZZ^*	$p_T^{4\ell}, p_T^j, \eta_j, D_{ZZ^*}$ $\Delta R_{4\ell, j}, \eta_{4\ell}, E_T^{\text{miss}}$	p_T^ℓ, η_ℓ	–	$\text{NN}_{\text{ggF+VBF}}$ for $\text{NN}_{ZZ} < 0.2$ NN_{ZZ} for $\text{NN}_{ZZ} > 0.2$
1j- $p_T^{4\ell}$ -High	ggF, VBF	$p_T^{4\ell}, p_T^j, \eta_j, D_{ZZ^*}$ $\Delta R_{4\ell, j}, \eta_{4\ell}, E_T^{\text{miss}}$	p_T^ℓ, η_ℓ	–	NN_{ggF}
2j	ggF, VBF, VH	$m_{jj}, p_T^{4\ell jj}, \eta_{ZZ}^{\text{zepp}}$	p_T^ℓ, η_ℓ	p_T^j, η_j, m_j	NN_{ggF} for $\text{NN}_{VH} < 0.3, \text{NN}_{\text{VBF}} < 0.1$ NN_{VBF} for $\text{NN}_{VH} < 0.3, \text{NN}_{\text{VBF}} > 0.1$ NN_{VH} for $\text{NN}_{VH} > 0.3, \text{NN}_{\text{VBF}} < 0.1$
2j-BSM-like	ggF, VBF	$m_{jj}, p_T^{4\ell jj}, \eta_{ZZ}^{\text{zepp}}$	p_T^ℓ, η_ℓ	p_T^j, η_j, m_j	NN_{VBF}

approximation [129] is validated with pseudo-experiments; where fewer than three events are observed in a bin, the quoted uncertainties are corrected to those obtained from pseudo-experiments.

The compatibility of a measured cross-section and its theoretical prediction is tested by computing a p -value based on the difference between the value of $-2 \ln \Lambda$ at the best-fit value and the value obtained by fixing the cross-section in each bin to that predicted by SM theory. These p -values do not include the uncertainties in the theoretical predictions.

7 Systematic uncertainties

The dominant uncertainty in the measurement arises from limited data statistics. In addition, a number of systematic uncertainties are considered. Experimental uncertainties arise from object reconstruction, identification, isolation, energy and momentum scale, resolution, and trigger efficiencies, the determination of the integrated luminosity and the data-driven background estimation. Additional theoretical uncertainties are related to the modelling of the signal and background processes.

7.1 Experimental uncertainties

The uncertainty in the integrated luminosity is 1.9% [31]. This affects the signal and background yields predicted by simulation if no normalisation to data is used as well as the extracted cross-section values. The uncertainty due to pile-up modeling are below 2% in the entire $m_{4\ell}$ range considered.

The reconstruction and identification efficiencies for muons and electrons are determined using the methods described in Refs. [109, 112]. For muons, these have associated uncertainties between 0.1% and 1.3%. For electrons with $p_T < 15$ GeV conservative uncertainties up to 17% are included to cover the change in detector conditions between Runs 2 and 3 and updates to the simulation. The isolation efficiency uncertainty is approximately 1% for electrons and 0.5 – 2% for muons.

Lepton energy and momentum scale and resolution are corrected to describe the data using the procedures documented in Refs. [111, 113], and the associated uncertainties have an impact at the most of 1 – 1.5% on the measured cross-sections.

Jet energy scale and resolution are calibrated using the procedure described in Ref. [116]. An additional uncertainty is assigned to jets in the kinematic region of $|\eta| > 3.2$ to account for an observed mismodeling on the jet energy calibration in the forward calorimeters.

Uncertainties on the b -jet identification efficiency are taken from 2022 and 2023 data and applied also to 2024 data and conservatively increased by ten times to reflect possible differences between different data taking periods. This is done only in the $t\bar{t}H$ category and tXX sideband, where b -jets are required and the impact is at the level of 1%.

Uncertainties on the lepton trigger efficiency have a negligible impact on the analysis, as does the uncertainty on the measured Higgs boson mass of $m_H = 125.09 \pm 0.24$ GeV [14].

For the data-driven measurement of the reducible background, several sources of uncertainty are considered. The dominant contribution arises from the statistical uncertainty in the region containing a quadruplet with at least one baseline lepton failing the tight requirements, where the fake factors are applied. An additional significant contribution comes from the systematic uncertainty associated with the choice of control regions used to derive the fake factors. This uncertainty reflects the variation of the result when the fake factors are extracted using only CR_Z or only CR_{top} , instead of their nominal combination, as described in Section 5. Those uncertainties have an impact that ranges from 1% to 9% both in the differential and in the production cross-section measurements. The statistical uncertainty due to the limited statistics in the CRs is also propagated to the fake factor estimation and has a negligible impact on the final yields. Finally, the uncertainty in the WZ normalisation estimated varying the normalisation scale factors is also negligible.

7.2 Theoretical uncertainties

Uncertainties in the signal prediction due to missing higher-order QCD effects are estimated using the scheme described in Ref. [130]. Parton shower and multiple-parton interaction modelling uncertainties on the acceptance are estimated using a prediction based on HERWIG7 as alternative parton shower generator and using tune eigenvector variations within the nominal parton shower generator tune, A14. Uncertainties related to the PDF set are evaluated using 40 eigenvector variations of the PDF4LHC21 PDF set. An additional uncertainty of the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay branching ratio of 2.2% [84, 131] is assigned when extrapolating to the total phase-space. These theory-related uncertainties enter the measurements in

different ways. When measuring the cross-section for both measurements, these uncertainties affect the result mainly through acceptance and migration effects and are generally small. In contrast, for the global strength measurement and interpretation of the results, the parameters of interest (μ) are defined relative to the SM predictions. In this case the same modelling uncertainties propagate directly to the predicted signal yields and have a larger impact.

Uncertainties in the prediction of the ZZ^* background from missing higher-order QCD effects are taken into account by varying the factorisation and renormalisation scales in the calculation by factors of two and choosing the configuration with the largest impact. PDF-related uncertainties are evaluated using the alternative PDF of the NNPDF3.0 PDF set [96]. A further modeling uncertainty is assigned by comparing the prediction from the nominal SHERPA sample to a simulation in MADGRAPH5_AMC@NLO matched to PYTHIA 8 with the FxFx prescription [44, 49]. As this background contribution is normalised to data, only the change in differential shapes impacts the measurements ($1 - 3\%$), while the effect in the shapes of the NN discriminants and in the relative contribution of this background between the sidebands and the signal regions for the STXS measurement is negligible. For the tXX process, uncertainties due to PDF and missing higher-order QCD effects are estimated from simulation using a similar procedure as the ZZ^* process.

Finally, unfolding-related systematics for the inclusive and differential cross-section measurements arise from uncertainties in the production-mode composition, that affect the response matrices. Additional uncertainties come from a potential bias introduced by the unfolding method. This bias is evaluated independently in each bin by comparing the unfolded cross-section in simulation with the expected result when the underlying true cross-sections of the simulated sample are varied within their expected statistical uncertainty. The impact of these uncertainties is negligible.

8 Results

Figure 4 shows the expected and observed four-lepton invariant mass distribution $m_{4\ell}$ for all Higgs boson candidates accepted by the event selection. The observed number of events and the expected signal and background yields in the mass range $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$ are presented in Table 4 before fitting to the data (pre-fit). The data agree with predictions.

Table 4: Expected (pre-fit) and observed numbers of events in the four decay final states after the event selection, in the mass range $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$. The column ‘Other Backgrounds’ includes the reducible background and other minor sources discussed in Section 5. The sum of the expected number of SM Higgs boson events and the estimated background yield is compared with the data. Combined statistical and systematic uncertainties are included for the predictions (see Section 7).

Final state	Signal	ZZ^*	Other Backgrounds	Total SM	Observed
4μ	90.5 ± 5.3	41.2 ± 1.6	4.5 ± 0.3	136 ± 6	136
$2e2\mu$	63.4 ± 4.0	30.0 ± 1.3	4.7 ± 0.5	98 ± 5	98
$2\mu2e$	40.7 ± 9.1	15.6 ± 4.4	5.6 ± 0.2	62 ± 13	67
$4e$	36.5 ± 9.2	14.1 ± 4.3	5.3 ± 0.3	56 ± 13	59
Total	231 ± 23	101 ± 10	20.1 ± 1.3	352 ± 32	360

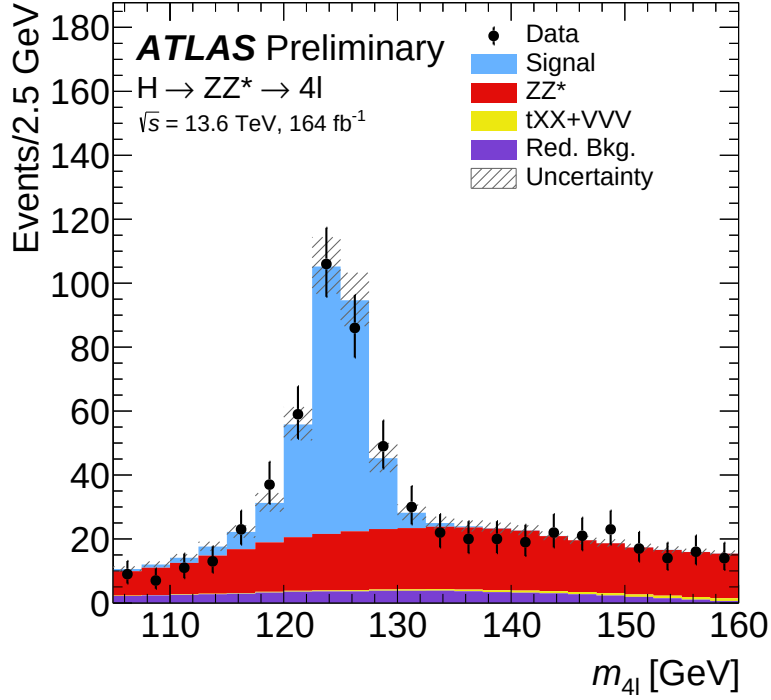


Figure 4: The observed and expected (pre-fit) inclusive four-lepton invariant mass distributions for the selected Higgs boson candidates. The uncertainty in the predictions is shown by the hatched band, which includes the theoretical uncertainties of the SM cross-section for the signal and the background processes.

8.1 Inclusive fiducial and total cross-sections

The inclusive fiducial and total cross-sections of the $H \rightarrow ZZ^* \rightarrow 4\ell$ process are measured by combining the results from the four decay final states assuming the relative SM branching ratio and are summarised in Figure 5. A normalisation factor of $\mu_{ZZ} = 0.94 \pm 0.08$ is obtained for the ZZ^* background component. The fiducial cross-section is measured as

$$\sigma_{\text{fid}} \equiv \sigma \cdot \mathcal{B}(H \rightarrow ZZ^* \rightarrow 4\ell) = 3.65 \pm 0.30(\text{stat.}) \pm 0.18^{+0.18}_{-0.16}(\text{sys.}) \text{ fb} = 3.65^{+0.35}_{-0.33} \text{ fb},$$

consistent with the SM prediction of $\sigma_{\text{fid}}^{\text{SM}} = 3.68 \pm 0.17 \text{ fb}$. The resulting total cross-section is

$$\sigma_{\text{total}} = 59.3 \pm 4.8(\text{stat.}) \pm 3.1(\text{sys.}) \text{ pb} = 59.3 \pm 5.7 \text{ pb},$$

consistent with the SM value of $\sigma_{\text{total}}^{\text{SM}} = 59.9 \pm 3.0 \text{ pb}$. The systematic uncertainty in the measurement is dominated by lepton reconstruction and isolation efficiencies. Larger uncertainties in decay channels with electrons in the subleading lepton pair reflect the conservative estimates of uncertainties on electron reconstruction efficiency. In combination with other final states, the impact of this systematics is reduced in the inclusive fiducial measurement. The probability of compatibility of the measured fiducial cross-section and the SM expectation is at the level of 92% and for the individual channels it ranges from 75% for $2\mu 2e$ to 90% for $4e$ final states.

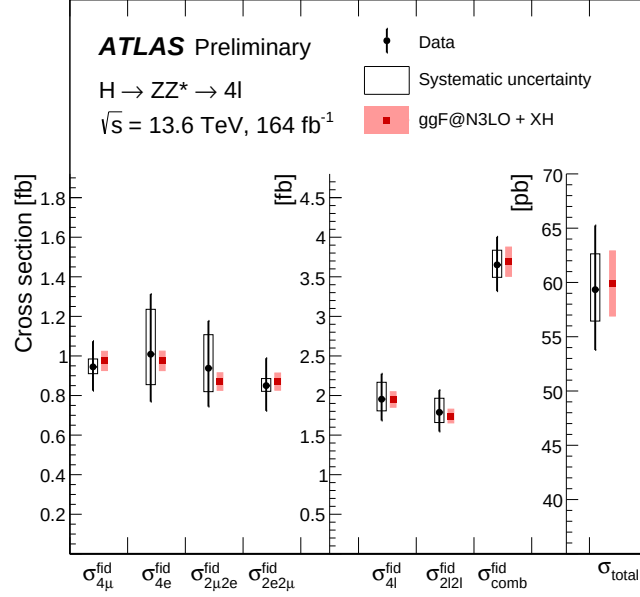
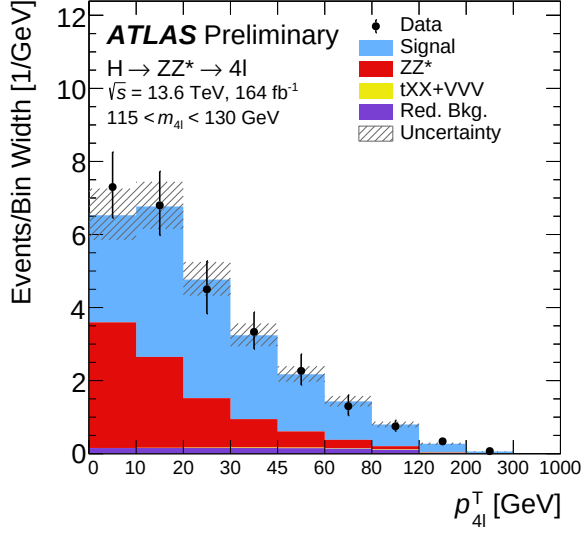


Figure 5: Measured fiducial and total cross-sections: (left panel) fiducial cross-sections for the four individual final states; (middle panel) inclusive fiducial cross-section obtained by combining all 4ℓ final states assuming relative SM branching ratios; (right panel) total cross-section obtained after accounting for the branching ratio of the Higgs boson into four leptons and the fiducial acceptance as determined from the simulated SM Higgs boson samples. The measurement in data is shown as black markers with error bands indicating the total and boxes indicating the systematic uncertainty. The measurements are compared to theoretical predictions with their associated uncertainties.

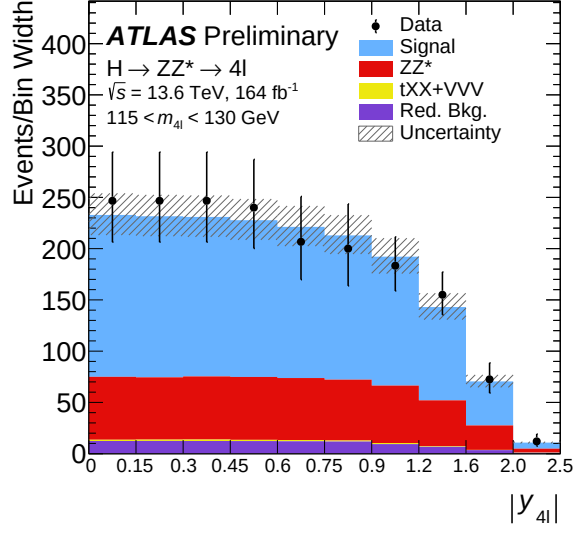
8.2 Differential cross-sections

Figure 6 and Figure 7 show the expected (pre-fit) and observed distributions of the $p_T^{4\ell}$, $y_{4\ell}$, m_{12} , m_{34} , N_{jets} , $\Delta\phi_{jj}$, ϕ observables used for the differential cross-section measurements, and the $p_T^{4\ell}$ vs. m_{34} distribution used for the double-differential cross-section measurement. The data are found to be largely in agreement with the prediction.

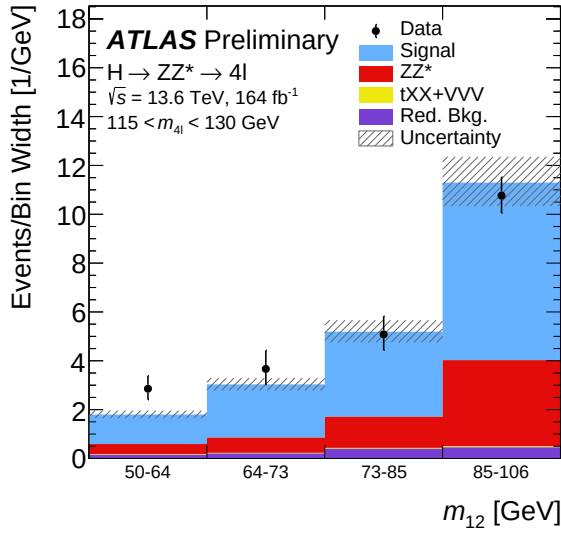
The corresponding measured differential cross-sections obtained from these distributions using the procedure discussed in Section 6.1 as well as the corresponding normalisation factors for the background are depicted in Figure 8 and Figure 9. Uncertainties are statistically dominated, and the subdominant systematic component arises primarily from lepton efficiency and (for jet-related observables) jet energy calibration uncertainties. The data are compared with SM expectations constructed from the ggF predictions provided by POWHEG and MADGRAPH5_AMC@NLO, which are normalised to N³LO cross-section as described in Section 3.2.2, with k-factors of 1.1 and 1.5 respectively. The uncertainty for POWHEG predictions is estimated from the procedure described in Section 7, while those for MADGRAPH5_AMC@NLO are estimated from varying factorisation and renormalisation scales. The probability of compatibility of the measured differential cross-section and the SM expectation ranges from 5% to approximately 100%, with the m_{12} being the least compatible with SM predictions due to a fluctuation in the first bin of the distribution.



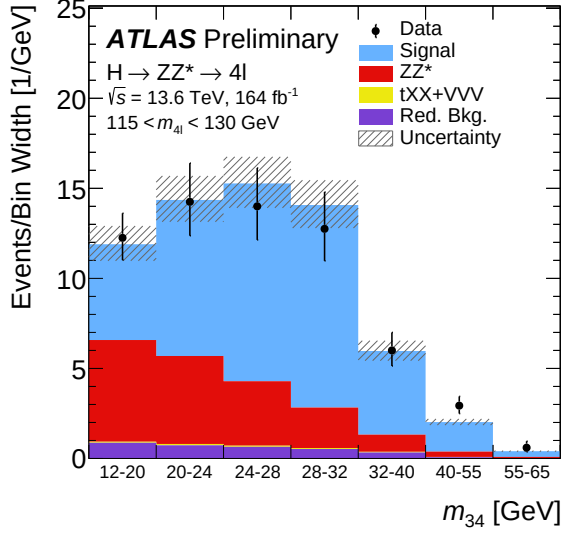
(a) $p_T^{4\ell}$



(b) $|y_{4\ell}|$

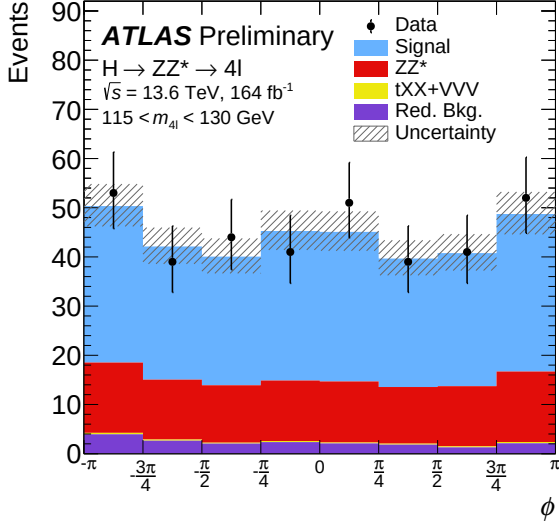


(c) m_{12}

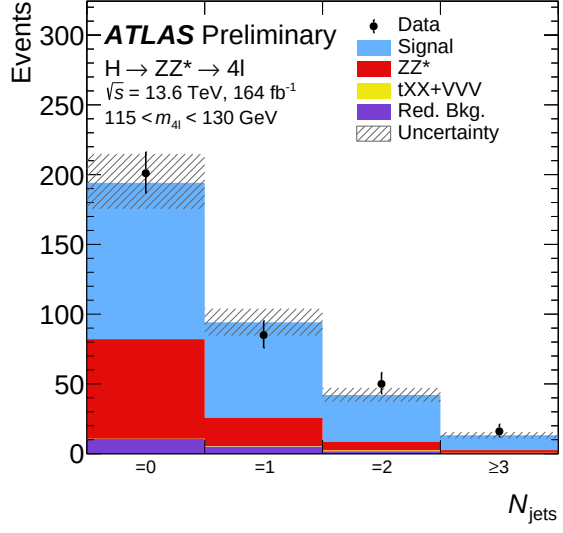


(d) m_{34}

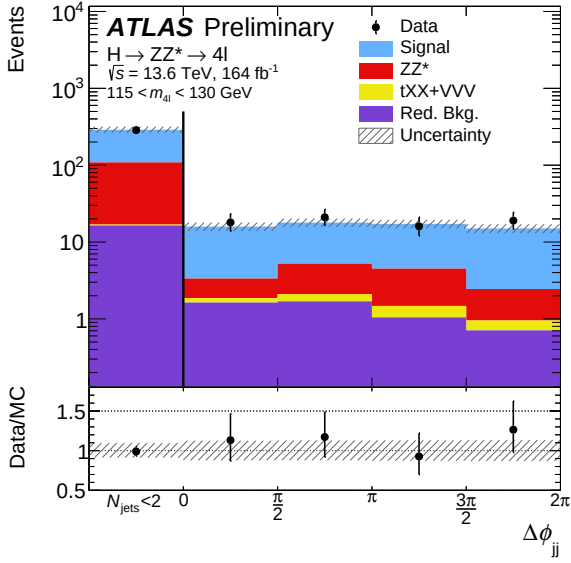
Figure 6: The observed and expected (pre-fit) distributions of (a) $p_T^{4\ell}$, (b) $|y_{4\ell}|$, (c) m_{12} , and (d) m_{34} observables, after the event selection in the mass range $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$. The uncertainty in the predictions is shown by the hatched band, which includes the theoretical uncertainties of the SM cross-section for the signal and the background processes.



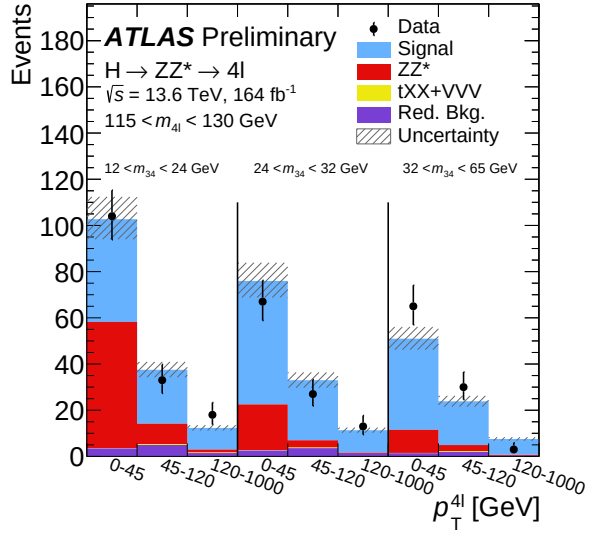
(a) ϕ



(b) N_{jets}



(c) $\Delta\phi_{jj}$



(d) $p_T^{4\ell}$ vs. m_{34}

Figure 7: The observed and expected (pre-fit) distributions of (a) ϕ , (b) N_{jets} , (c) $\Delta\phi_{jj}$ and (d) $p_T^{4\ell}$ vs. m_{34} observables, after the event selection in the mass range $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$. The uncertainty in the predictions is shown by the hatched band, which includes the theoretical uncertainties of the SM cross-section for the signal and the background processes.

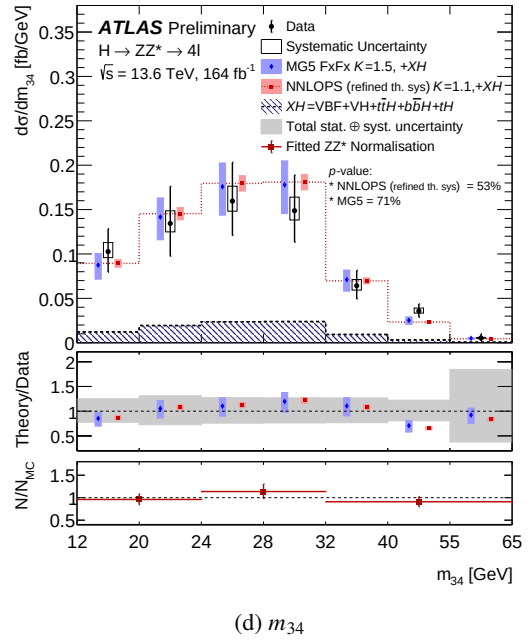
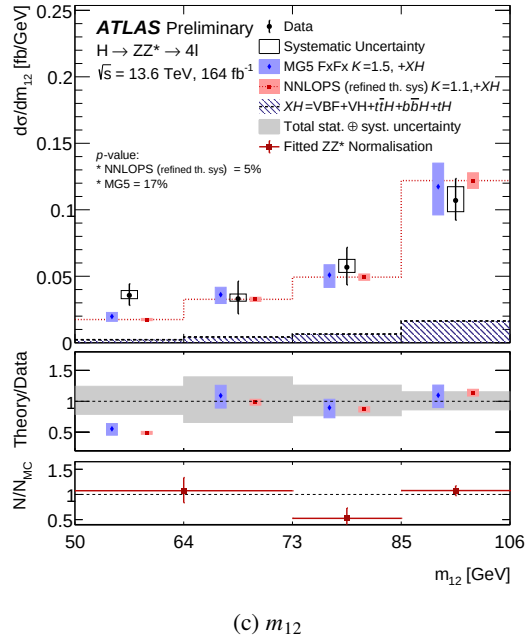
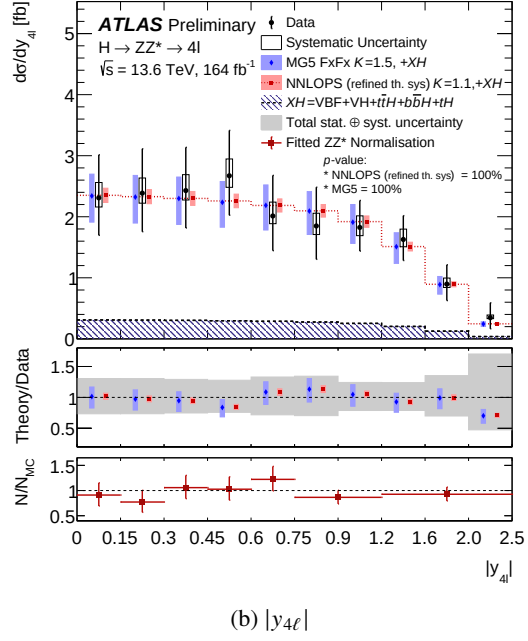
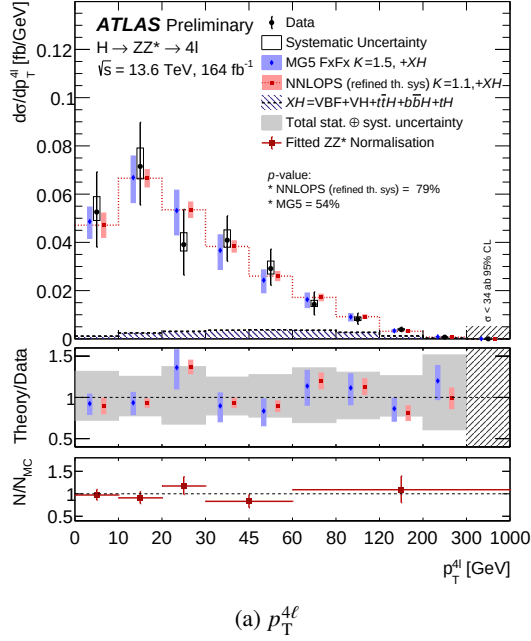


Figure 8: Differential fiducial cross-sections as a function of (a) $p_T^{4\ell}$, (b) $|y_{4\ell}|$, (c) m_{12} , and (d) m_{34} . The upper panel shows the measured cross-sections, compared with ggF predictions by POWHEG (NNLOPS) and MADGRAPH5_AMC@NLO (MG5), which are normalised to the $N^3\text{LO}$ total cross-section with the listed K -factors. The shaded bands on the expected cross sections indicate the PDF and scale systematic uncertainties, in particular the ones for NNLOPS are calculated as described in [130], while for MG5 are estimated from varying factorisation and renormalisation scales. The markers denoting the simulation are horizontally displaced relative to the measurement in data to improve readability. MC-based predictions for all other Higgs boson production-modes XH are normalised to the SM predictions. The error bars on the data points show the total uncertainties, while the systematic uncertainties are indicated by the boxes. The middle panel shows the ratio of different predictions to the data, and the grey area represents the total uncertainty of the measurement. The bottom panel shows the ratios of the fitted values of the ZZ^* normalisation factors to the predictions from MC simulation.

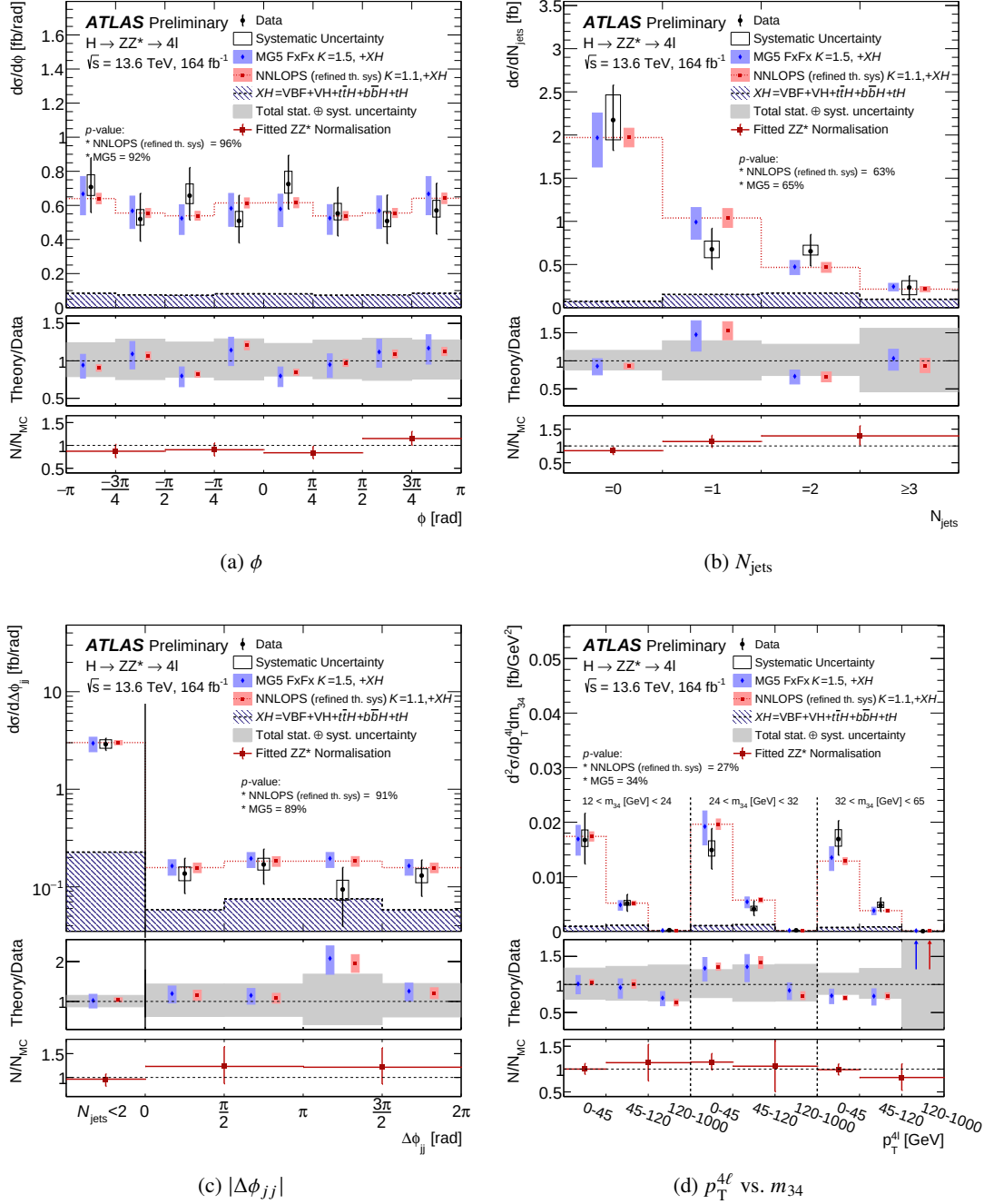


Figure 9: Differential fiducial cross-sections as a function of (a) ϕ , (b) N_{jets} , (c) $\Delta\phi_{jj}$ and (d) $p_T^{4\ell}$ vs. m_{34} . The upper panel shows the measured cross-sections, compared with ggF predictions by POWHEG (NNLOPS) and MADGRAPH5_AMC@NLO (MG5), which are normalised to the N³LO total cross-section with the listed K -factors. The shaded bands on the expected cross sections indicate the PDF and scale systematic uncertainties, in particular the ones for NNLOPS are calculated as described in [130], while for MG5 are estimated from varying factorisation and renormalisation scales. The markers denoting the simulation are horizontally displaced relative to the measurement in data to improve readability. MC-based predictions for all other Higgs boson production-modes XH are normalised to the SM predictions. The error bars on the data points show the total uncertainties, while the systematic uncertainties are indicated by the boxes. The middle panel shows the ratio of different predictions to the data, and the grey area represents the total uncertainty of the measurement. The bottom panel shows the ratios of the fitted values of the ZZ^* normalisation factors to the predictions from MC simulation.

8.3 Cross-sections per production-mode

For the production cross-section measurement, the expected numbers of signal and background events in each reconstructed event category are shown in Table 5 together with the corresponding observed number of events. They are in good agreement. The observed and expected (pre-fit) distributions of the transverse momenta of the leading lepton, the kinematic discriminant D_{ZZ^*} , the transverse momenta of the leading jet and the angular separation between the four lepton system and the leading jet $\Delta R_{4\ell,j}$, which are used as inputs variable for the NN training in the reconstructed categories, are shown in Figure 10 and are in good agreement as well.

Table 5: The expected (pre-fit) and observed numbers of events are shown for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$. Results are presented for the signal region $115 < m_{4\ell} < 130 \text{ GeV}$ and the sideband regions $105 < m_{4\ell} < 115 \text{ GeV}$ and $130 < m_{4\ell} < 160 \text{ GeV}$ (350 GeV for tXX -enriched) in each reconstructed event category, assuming a SM Higgs boson signal with a mass $m_H = 125 \text{ GeV}$. The sum of the expected SM Higgs boson signal and the estimated background yields is compared with the data. Combined statistical and systematic uncertainties are included for the predictions. Expected contributions below 0.2% of the total yield in each reconstructed event category are not shown and are replaced by ‘-’.

Reconstructed event category	Signal	ZZ^* background	tXX background	Other backgrounds	Total expected	Observed
Signal $115 < m_{4\ell} < 130 \text{ GeV}$						
$0j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$	26 ± 5	31.4 ± 3.2	-	1.750 ± 0.035	59 ± 7	64
$0j\text{-}p_{\text{T}}^{4\ell}\text{-Med}$	85 ± 11	40 ± 4	-	5.86 ± 0.26	131 ± 15	135
$0j\text{-}p_{\text{T}}^{4\ell}\text{-High}$	0.38 ± 0.05	0.041 ± 0.007	0.0243 ± 0.0005	0.149 ± 0.014	0.60 ± 0.06	1
$1j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$	38 ± 5	15.3 ± 2.1	0.1120 ± 0.0021	3.21 ± 0.33	57 ± 7	54
$1j\text{-}p_{\text{T}}^{4\ell}\text{-Med}$	23.2 ± 3.4	4.3 ± 0.4	0.1380 ± 0.0026	1.69 ± 0.18	29 ± 4	24
$1j\text{-}p_{\text{T}}^{4\ell}\text{-High}$	5.4 ± 0.9	0.56 ± 0.05	0.0471 ± 0.0009	0.2861 ± 0.0013	6.3 ± 0.9	7
$1j\text{-}p_{\text{T}}^{4\ell}\text{-BSM-Like}$	1.4 ± 0.4	0.083 ± 0.007	0.00761 ± 0.00014	0.030 ± 0.009	1.5 ± 0.4	0
$2j$	44 ± 7	8.9 ± 1.4	0.956 ± 0.018	4.07 ± 0.33	58 ± 8	67
$2j\text{-BSM-like}$	4.1 ± 0.8	0.197 ± 0.013	0.0333 ± 0.0006	0.58 ± 0.06	4.9 ± 0.8	4
$VH\text{-Lep-enriched}$	1.42 ± 0.12	0.256 ± 0.029	0.0591 ± 0.0011	0.365 ± 0.018	2.10 ± 0.15	2
$ttH\text{-enriched}$	1.68 ± 0.20	0.037 ± 0.008	0.292 ± 0.006	0.301 ± 0.027	2.31 ± 0.20	2
Sideband $105 < m_{4\ell} < 115 \text{ GeV}$ or $130 < m_{4\ell} < 160 \text{ GeV}$						
SB- $0j$	5.9 ± 1.0	161 ± 15	-	21.06 ± 0.35	189 ± 16	153
SB- $1j$	3.8 ± 0.6	53 ± 5	1.644 ± 0.031	12.9 ± 0.4	71 ± 6	74
SB- $2j$	3.0 ± 0.5	26 ± 4	4.44 ± 0.08	7.237 ± 0.020	40 ± 4	48
Sideband $105 < m_{4\ell} < 115 \text{ GeV}$ or $130 < m_{4\ell} < 350 \text{ GeV}$						
SB- $tXX\text{-enriched}$	0.096 ± 0.012	0.27 ± 0.07	5.29 ± 0.10	2.29 ± 0.14	7.95 ± 0.19	15

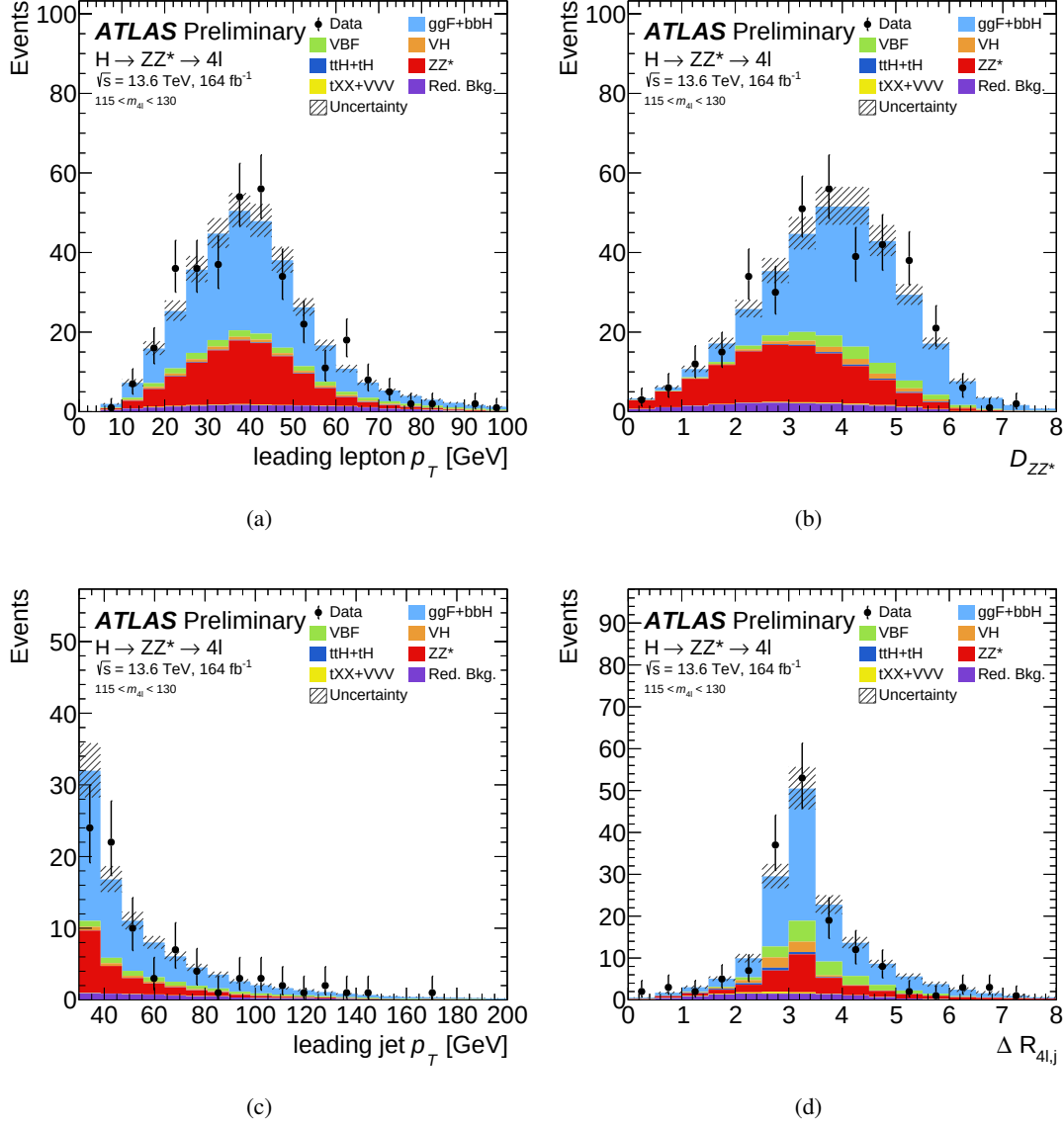


Figure 10: The observed and expected distributions (pre-fit) of (a) the transverse momenta of the leading lepton (b) the kinematic discriminant D_{ZZ^*} (c) the transverse momenta of the leading jet and (d) angular separation between the four lepton system and the leading jet $\Delta R_{4l,j}$. The SM Higgs boson signal is assumed to have a mass $m_H = 125$ GeV. The uncertainty in the prediction is shown by the hatched band, calculated as described in Section 7.

The observed and expected (post-fit) distributions of the NN discriminants are shown in Figure 11 and Figure 12. The $1j\text{-}p_T^{4\ell}\text{-Low}$ category is additionally split into two sub-categories with NN_{ZZ} greater or smaller than 0.2 to target different sub-processes as defined in Table 3. Similarly, the $2j$ is divided into three sub-categories with $(NN_{VH} < 0.3, NN_{VBF} < 0.1)$, $(NN_{VH} < 0.3, NN_{VBF} > 0.1)$ and $(NN_{VH} > 0.3, NN_{VBF} < 0.1)$ to target the ggF, VBF and VH processes. In addition, Figure 12(e) and Figure 12(f) show the observed and expected yields in the categories where no NN discriminant is used and in the mass sidebands used to constrain the ZZ^* and tXX background, respectively. All distributions are in good agreement with the data.

The observed values of $\sigma_{|y_H| < 2.5} \cdot \mathcal{B}(H \rightarrow ZZ^*)$ in the Higgs boson rapidity range of $|y_H| < 2.5$, the corresponding expected SM cross-sections, and their ratios are shown in Figure 13 for the inclusive production and for the individual production-modes as defined in Section 6.2. Similarly, the results for the Reduced Stage 1.2 bins are shown in Figure 14. The branching ratio for the $ZZ^* \rightarrow 4\ell$ decay is fixed to the SM prediction. In the ratio calculation, uncertainties on the SM expectation are not taken into account. An inclusive cross-section of

$$\sigma_{|y_H| < 2.5} \cdot \mathcal{B}(H \rightarrow ZZ^*) = 1.42 \pm 0.11(\text{stat.}) \pm 0.12(\text{sys.}) \text{ pb} = 1.42 \pm 0.16 \text{ pb}$$

is measured, in agreement with the SM prediction of $\sigma_{|y_H| < 2.5}^{\text{SM}} \cdot \mathcal{B}^{\text{SM}}(H \rightarrow ZZ^*) = 1.44 \pm 0.08 \text{ pb}$. Normalisation factors for the ZZ^* continuum background of $N_{ZZ}^{\text{SB-}0j}/N_{\text{MC}} = 0.82 \pm 0.14$, $N_{ZZ}^{\text{SB-}1j}/N_{\text{MC}} = 1.18^{+0.22}_{-0.19}$, and $N_{ZZ}^{\text{SB-}2j}/N_{\text{MC}} = 1.53^{+0.46}_{-0.37}$ are obtained from the sidebands, separately for the three event categories with different jet multiplicities, while the one for the tXX backgrounds from the specific category $N_{tXX}^{\text{SB-}tXX}/N_{\text{MC}} = 2.5^{+0.9}_{-0.7}$.

The data are also interpreted in terms of the global signal strength, where the theoretical uncertainties in the SM cross-section predictions are included in the measurement, yielding

$$\mu = 0.99 \pm 0.08(\text{stat.}) \pm 0.10(\text{sys.}) = 0.99 \pm 0.13$$

The dominant systematic uncertainty in inclusive cross-section measurement is the experimental uncertainty in the lepton efficiency and the integrated luminosity measurements. For the global strength measurement theoretical uncertainties related to higher order correction on ggF corrections are dominant.

The dominant contribution to the measurement uncertainty in the ggF production-mode originates from the same sources as in the inclusive measurement, as well as in the Reduced Stage 1.2 bins with zero jets. For the VBF production bin, the theoretical systematic uncertainties related to missing higher-order QCD corrections and the experimental one on the jet energy scale and resolution uncertainties become relevant, as well as in the VH and $t\bar{t}H$ production-modes. For the Reduced Stage 1.2 bins with higher jet multiplicity the dominant cross-section uncertainties are the jet energy scale and resolution, and missing higher-order corrections. The probability of compatibility of the measured production-mode (Reduced Stage 1.2) measurement and the SM expectation is 51% (68%).

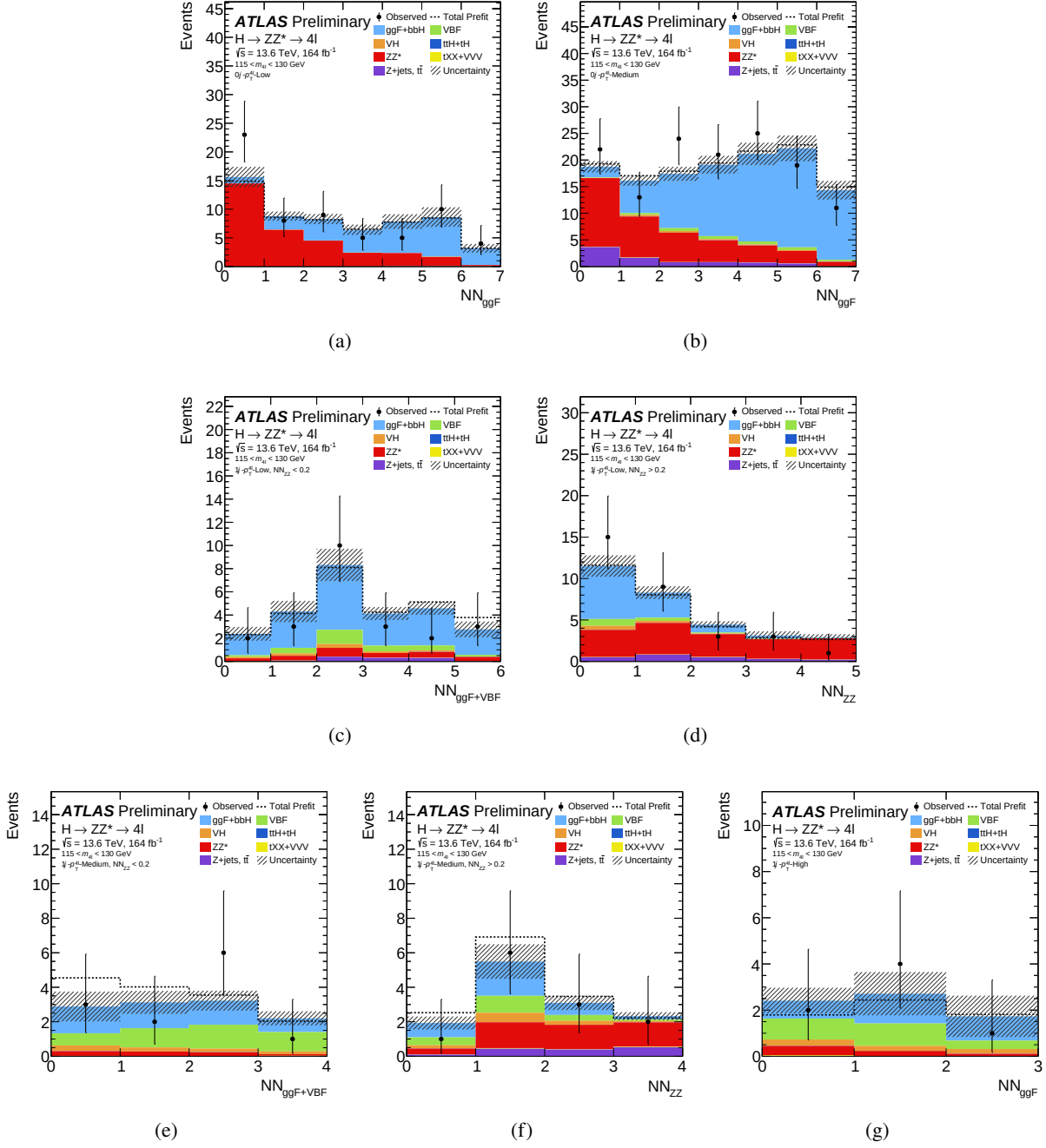


Figure 11: The observed and expected neural network output (post-fit) distributions for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6$ TeV in the different reconstructed event categories: (a) NN_{ggF} in $0j-p_T^{4\ell}$ -Low, (b) NN_{ggF} in $0j-p_T^{4\ell}$ -Med, (c) $NN_{ggF+VBF}$ and (d) NN_{ZZ} in $1j-p_T^{4\ell}$ -Low for low and high NN_{ZZ} , respectively, (e) $NN_{ggF+VBF}$ and (f) NN_{ZZ} in $1j-p_T^{4\ell}$ -Med for low and high NN_{ZZ} , respectively and (g) NN_{ggF} in $1j-p_T^{4\ell}$ -High. The total post-fit distribution is shown as a stacked histogram, while the total pre-fit distribution is shown with dotted black line for comparison. The uncertainty in the prediction, comprising the theoretical uncertainties in the SM cross-section for the signal and the background processes, is shown by the hatched band. The bin boundaries are chosen to maximise the significance of the targeted signal in each category. The individual bins do not map linearly to discriminant values.

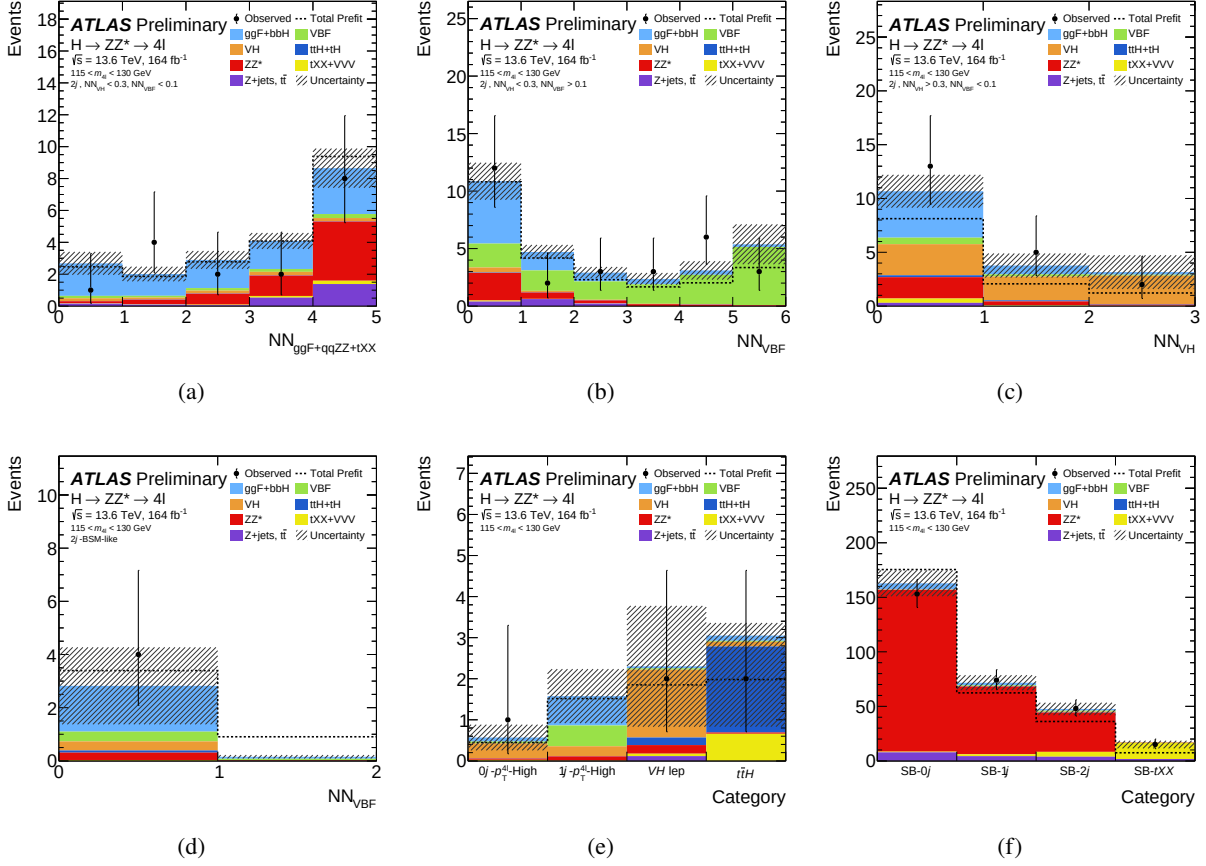


Figure 12: The observed and expected neural network output (post-fit) distributions for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$ in the different reconstructed event categories: (a) NN_{ggF} (b) NN_{VBF} and (c) NN_{VBF} with the different selection on NN_{VBF} and NN_{VH} in $2j$ categories, (d) NN_{VBF} in $2j$ -BSM-like. (e) Shows the categories where no NN discriminant is used while (f) shows the sidebands used to constrain the ZZ^* and tXX backgrounds. The total post-fit distribution is shown as a stacked histogram, while the total pre-fit distribution is shown with dotted black line for comparison. The uncertainty in the prediction, comprising the theoretical uncertainties in the SM cross-section for the signal and the background processes, is shown by the hatched band. The bin boundaries are chosen to maximise the significance of the targeted signal in each category. The individual bins do not map linearly to discriminant values.

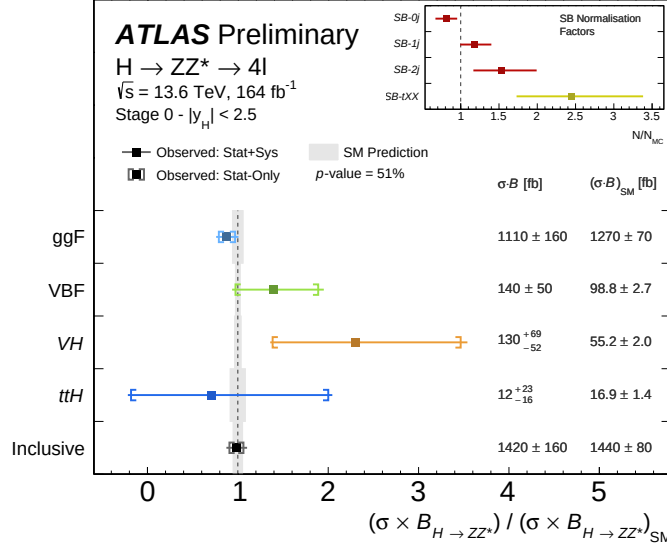


Figure 13: The observed and expected SM values of the inclusive and individual production-mode cross-sections $\sigma_{|y_H| < 2.5} \cdot \mathcal{B}(H \rightarrow ZZ^*)$ normalised by the SM expectation for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$. The branching ratio for the $ZZ^* \rightarrow 4\ell$ decay is assumed to be given by the SM prediction. The horizontal bands around the expected SM result indicate the theoretical uncertainty in the prediction.

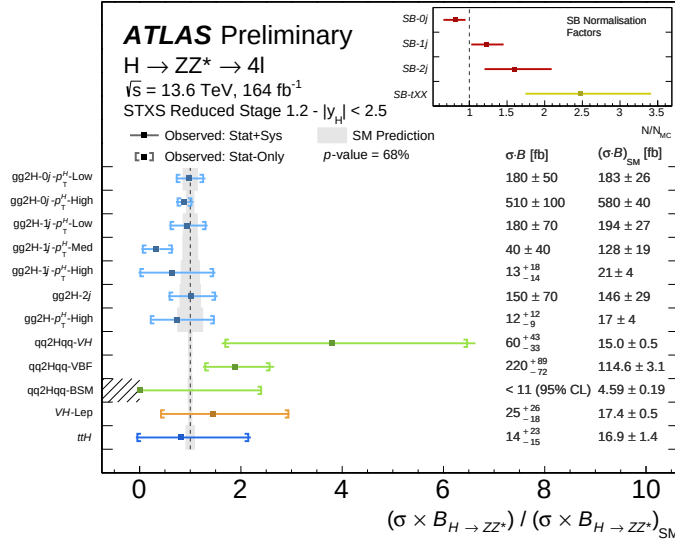


Figure 14: The observed and expected SM values of the inclusive and individual production-mode cross-sections $\sigma_{|y_H| < 2.5} \cdot \mathcal{B}(H \rightarrow ZZ^*)$ normalised by the SM expectation for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$. The branching ratio for the $ZZ^* \rightarrow 4\ell$ decay is assumed to be given by the SM prediction. The horizontal bands around the expected SM result indicate the theoretical uncertainty in the prediction. Due to zero observed events in the bins with highest sensitivity, an upper limit at 95%CL for the qq2Hqq-BSM bin is set. The hatched area indicates that the qq2Hqq-BSM parameter of interest is constrained to positive values in the fit.

9 Interpretations of the differential distributions and production cross-sections

9.1 Constraints on Higgs boson couplings within the κ -framework

To probe physics beyond the SM, the measured production-mode cross-sections are interpreted in the κ framework [132], which parametrises possible deviations of Higgs boson couplings from their SM values using coupling-strength modifiers $\tilde{\kappa}$. Under the narrow-width approximation, Higgs production and decay factorise, and $\sigma \cdot \mathcal{B}$ can be expressed in terms of modified Higgs production cross-sections and partial widths. The $\tilde{\kappa}$ dependence at $\sqrt{s} = 13.6$ TeV is obtained by linear interpolation between the corresponding terms at 13 and 14 TeV [35].

The interpretation is performed in a benchmark scenario with two universal modifiers: $\kappa_V = \kappa_W = \kappa_Z$ for Higgs-to-vector-boson couplings and $\kappa_f = \kappa_t = \kappa_b = \kappa_c = \kappa_\tau = \kappa_\mu$ for Higgs-to-fermion couplings, assuming a single Higgs boson state that is even under the combined charge-parity (CP) transformation, with the SM tensor structure and no undetected or invisible decays. For loop-induced processes, the resolved treatment is adopted, in which the effective loop couplings are expressed in terms of the modified couplings to the massive SM particles.

The results are statistically combined with the corresponding measurements performed with the full Run 2 data set [8], accounting for the systematic uncertainties described in Section 7. The following experimental uncertainties are treated as correlated between Run 2 and Run 3 results: the muon momentum scale and reconstruction efficiency; the electron energy scale and energy resolution; and the jet energy scale and resolution, except for Run 3-specific components. Other experimental uncertainties are treated as uncorrelated. Theoretical uncertainties from missing higher-order corrections affecting the Higgs boson signal cross-sections in exclusive kinematic regions, the Higgs boson mass uncertainty, and uncertainties in background estimates are treated as correlated. All other theoretical uncertainties, including PDF and parton shower uncertainties, are treated as uncorrelated.

The likelihood contours and best-fit values in the (κ_f, κ_V) plane are shown in Figure 15. Only the quadrant with $\kappa_f > 0$ and $\kappa_V > 0$ is displayed, as the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel has limited sensitivity to the relative sign of the two modifiers. The compatibility of the combined measurement with the SM expectation corresponds to a p -value of 29%.

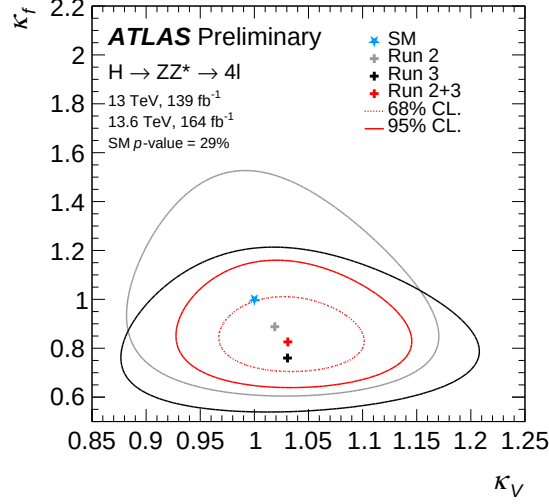


Figure 15: Likelihood contours in the (κ_f, κ_V) plane comparing the current results with those obtained in Run 2. The statistical combination of the two measurements is also presented. The area outside the solid (dashed) contours is excluded at 95% (68%) confidence level. The best-fit point (solid cross) is shown together with the SM prediction (star).

9.2 Constraints on CP-even and CP-odd effective field theory operators

The κ -framework assumes that the tensor structure of the Higgs boson couplings is the same as in the SM. To probe possible non-SM tensor structures, selected unfolded differential measurements are interpreted in an effective field theory (EFT) approach. In this approach, the BSM interactions are introduced via additional higher-dimensional operators $O_i^{(d)}$ of dimension d , supplementing the SM Lagrangian $\mathcal{L}_{\text{SM}}$,

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i^{(d)}}{\Lambda^{(d-4)}} O_i^{(d)} \quad \text{for } d > 4,$$

where $C_i^{(d)}$ are the Wilson coefficients and Λ denotes the characteristic scale of new physics. In this analysis, only dimension-six operators are considered. Dimension-five and dimension-seven operators are not included, as they violate lepton or baryon number conservation and are therefore not relevant in the present context. Although dimension-eight operators can interfere with the SM amplitudes at a similar order as the quadratic contributions from dimension-six operators, their effect is not considered. For energies below the scale Λ , the data are sensitive only to the combinations $c_i = C_i^{(d=6)}/\Lambda^2$, which are the quantities constrained in this analysis.

Constraints are derived within the Standard Model EFT (SMEFT) formalism [133] in the Warsaw basis [134]. Given the sensitivity in the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel, only three CP-even and three CP-odd operators modifying Higgs interactions with electroweak gauge bosons are probed as summarised in Table 6. Operators affecting only double-Higgs production, as well as operators modifying Higgs couplings to quarks and leptons, are not considered. The impact of these operators on the total Higgs boson width is neglected.

The EFT interpretation is performed using observables selected to enhance sensitivity to the targeted operators and to retain discrimination between the interference (linear) and BSM-squared (quadratic) contributions. The two-dimensional $(p_T^{4\ell}, m_{34})$ measurement is used to constrain c_{HW} , since it simultaneously

Table 6: Summary of CP-even and CP-odd EFT operators in the SMEFT formalism probed in the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel. The corresponding tensor structure in terms of the SM fields from Ref. [133] is shown together with the associated Wilson coefficients and the affected interaction vertices. $V_{\mu\nu}$ is the field-strength tensor for a given gauge field $V = W, B$.

Operator	Structure	Coeff.
O_{HW}	$HH^\dagger W_{\mu\nu}^l W^{\mu\nu l}$	c_{HW}
O_{HB}	$HH^\dagger B_{\mu\nu} B^{\mu\nu}$	c_{HB}
O_{HWB}	$HH^\dagger \tau^l W_{\mu\nu}^l B^{\mu\nu}$	c_{HWB}
$O_{H\tilde{W}}$	$HH^\dagger \tilde{W}_{\mu\nu}^l W^{\mu\nu l}$	$c_{H\tilde{W}}$
$O_{H\tilde{B}}$	$HH^\dagger \tilde{B}_{\mu\nu} B^{\mu\nu}$	$c_{H\tilde{B}}$
$O_{H\tilde{W}B}$	$HH^\dagger \tau^l \tilde{W}_{\mu\nu}^l B^{\mu\nu}$	$c_{H\tilde{W}B}$

probes production- and decay-side effects: $p_T^{4\ell}$ is sensitive to modifications of the production kinematics, while m_{34} provides sensitivity to changes in the decay matrix element in $H \rightarrow ZZ^* \rightarrow 4\ell$. The sensitivity of this approach depends on the chosen binning in m_{34} as the balance between statistical precision and shape sensitivity is required to be optimised.

For the remaining CP-even operators, the angular observable ϕ provides improved sensitivity compared with $(p_T^{4\ell}, m_{34})$. This improvement arises as the linear SMEFT contribution exhibits sign changes as a function of ϕ , leading to a periodic modulation that cancels when integrated over the full phase space. Consequently, the SM-BSM interference is suppressed in ϕ -inclusive observables, similarly to those observed in non-Higgs decays [135, 136]. Although ϕ is a CP-even observable, it also exhibits sensitivity to CP-odd operators, with the interference term showing an analogous behaviour to the CP-even case but with different periodicity. The $\Delta\phi_{jj}$ observable is also studied for CP-odd operators, but it does not provide discrimination between CP-even and CP-odd operators in the present dataset due to the dominant dependence on quadratic EFT terms.

The Wilson coefficients are constrained by comparing the unfolded signal cross-section in each bin of the chosen distribution to the SMEFT expectation, with the expected cross-section and acceptance parametrised using a matrix-element-based procedure [137]. For a bin j of a given observable, the expected cross-section is written as

$$\sigma_{pp \rightarrow H \rightarrow 4\ell, j}^{\text{SMEFT}}(c_i) = \sigma_{pp \rightarrow H \rightarrow 4\ell, j}^{\text{SM}} \cdot \sum_p \left(1 + \sum_i \alpha_i^{(p, j)} c_i + \sum_{i, k \geq i} \beta_{ik}^{(p, j)} c_i c_k \right) \cdot \frac{(1 + \sum_i \alpha_i^{(\Gamma_{4\ell, j})} c_i + \sum_{i, k \geq i} \beta_{ik}^{(\Gamma_{4\ell, j})} c_i c_k)}{(1 + \sum_i \alpha_i^{(\Gamma_{tot, j})} c_i + \sum_{i, k \geq i} \beta_{ik}^{(\Gamma_{tot, j})} c_i c_k)}, \quad (5)$$

where p labels Higgs boson production processes, and $\Gamma_{4\ell}$ and Γ_{tot} denote the partial width to four leptons and the total width, respectively. The terms linear in c_i encode SM-EFT interference, while the quadratic terms represent the pure dimension-six contribution. The coefficients α and β are obtained from matrix elements computed using dedicated simulated samples generated at LO in QCD with MADGRAPH5_AMC@NLO and the NNPDF23lo PDF set. The SMEFT contributions to VBF production and the HZZ decay vertex are modelled using SMEFTs_{SM} [133, 138]. The SMEFT impact on VH production is neglected, as it has a negligible effect on the considered distributions.

Since the measurements do not provide sufficient sensitivity to constrain the full set of coefficients simultaneously, fits are performed for one coefficient at a time, with all others set to zero. Two treatments

of the SMEFT expansion are presented: a *linear-only* interpretation retaining only the interference term, and a *linear+quadratic* interpretation including both interference and BSM-squared terms. Quadratic terms typically increase sensitivity through additional changes in total cross-section, and a comparison of the two treatments provides information on the impact of higher-order contributions and the possible onset of perturbative breakdown as quadratic effects become dominant. For each probed coefficient, one-dimensional likelihood scans are performed using the observable providing the best expected sensitivity: $(p_T^{4\ell}, m_{34})$ for c_{HW} and ϕ for rest. The observed and expected confidence intervals (CL) are summarised in Table 7, and the corresponding likelihood scans comparing the linear-only and linear+quadratic treatments are shown for the CP-even Wilson Coefficients in Figure 16. Additional results where combinations of two EFT operators are simultaneously probed are shown in Figure 17. For each combination, the scans are obtained from the observable which provides the best expected sensitivity. Two-dimensional EFT scans are not used to derive limits, as the likelihood exhibits poorly constrained directions in parameter space, leading to degeneracies and flat directions.

The corresponding limits on the new physics scale Λ for benchmark coefficient values are shown in Figure 18.

Table 7: Expected and observed confidence intervals (CL) at 95% CL on the CP-even SMEFT Wilson coefficients for an integrated luminosity of 164 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$, assuming $\Lambda = 1 \text{ TeV}$. Only one Wilson coefficient is fitted at a time while all others are set to zero. For each coefficient, the observable used in the fit is indicated, together with results obtained in the linear-only and linear+quadratic SMEFT treatments. $(p_T^{4\ell}, m_{34})$ observable is used for c_{HW} results, while ϕ is used for the other operators.

	Expected 95% CL [TeV^{-2}]		Observed 95% CL [TeV^{-2}]		Best fit [TeV^{-2}]	
	Linear	Linear + Quadratic	Linear	Linear + Quadratic	Linear	Linear + Quadratic
c_{HW}	$[-3.8, 3.2]$	$[-1.8, 1.8]$	$[-4.3, 2.9]$	$[-2.0, 1.8]$	-0.40	-0.08
c_{HB}	$[-0.63, 0.59]$	$[-0.54, 1.2]$	$[-0.58, 0.64]$	$[-0.49, 1.25]$	0.05	0.05
c_{HWB}	$[-1.5, 1.4]$	$[-0.42, 0.48]$	$[-1.6, 1.3]$	$[-0.40, 0.45]$	-0.08	-0.02
$c_{H\tilde{W}}$	$[-3.0, 3.0]$	$[-1.6, 1.6]$	$[-2.1, 3.9]$	$[-1.3, 1.6]$	0.88	0.38
$c_{H\tilde{B}}$	$[-1.5, 1.5]$	$[-1.0, 0.96]$	$[-1.1, 1.6]$	$[-0.80, 0.96]$	0.39	0.20
$c_{H\tilde{W}B}$	$[-2.3, 2.3]$	$[-0.45, 0.45]$	$[-3.1, 1.5]$	$[-0.44, 0.41]$	-0.80	-0.09

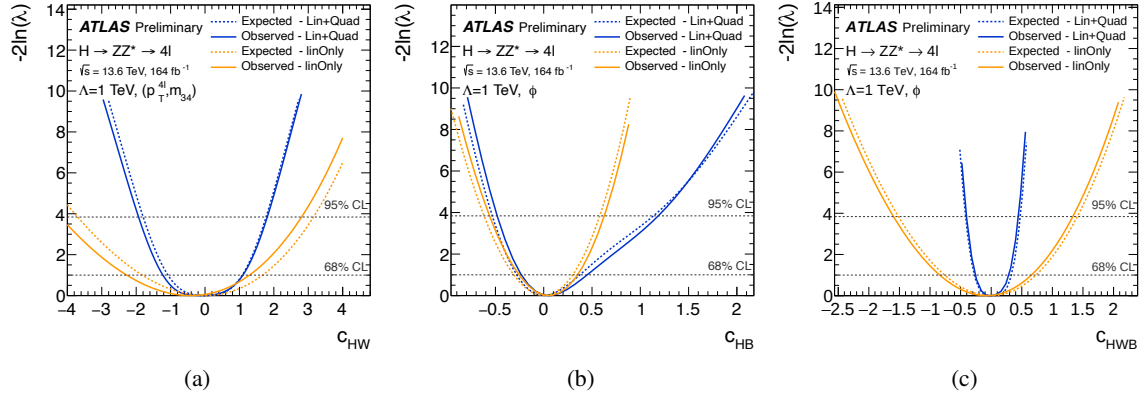


Figure 16: One-dimensional likelihood scans for the CP-even SMEFT Wilson coefficients, comparing the linear-only and linear+quadratic SMEFT treatments. The c_{HW} scan uses the unfolded ($p_T^{4\ell}, m_{34}$) measurement, while the other scans use the unfolded ϕ measurement. A new physics scale of $\Lambda = 1$ TeV is assumed.

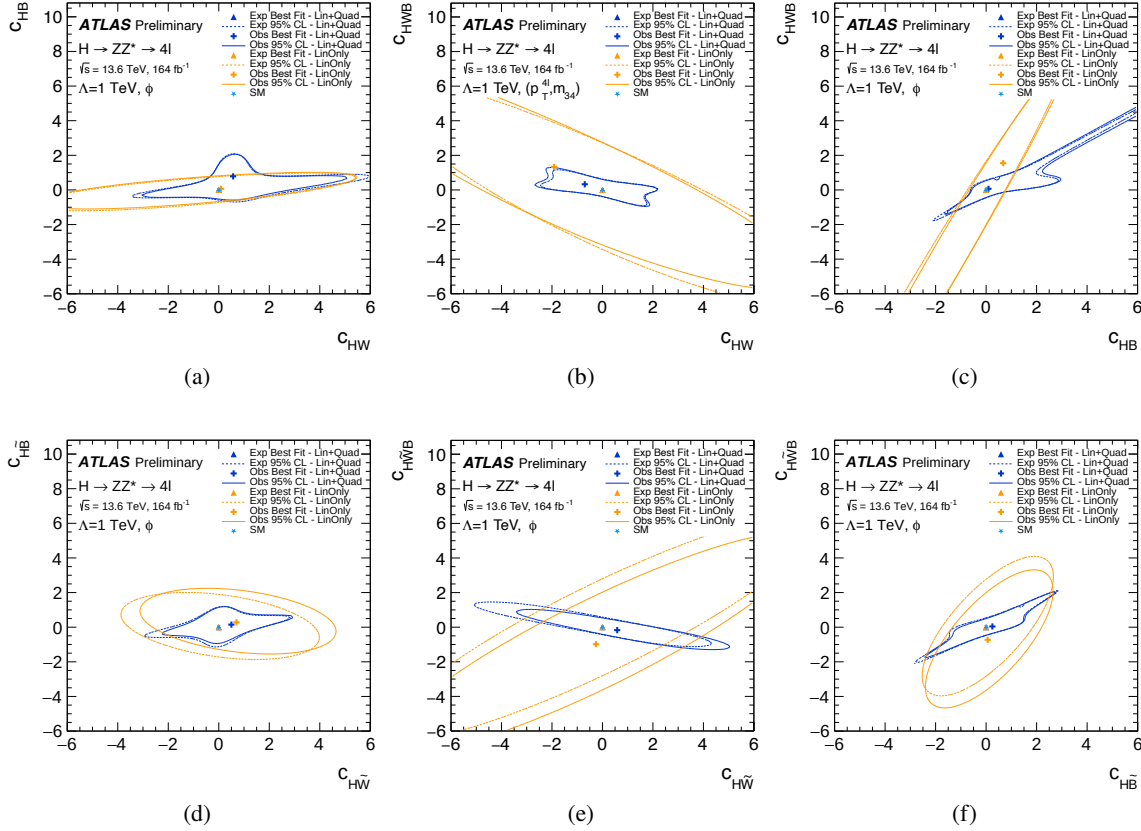


Figure 17: Expected (dashed line) and observed (full line) two-dimensional likelihood scans for the SMEFT Wilson coefficients, comparing the linear-only and linear+quadratic SMEFT treatments. Coupling values outside the solid lines are excluded at 95% confidence level (CL). The best fit to the data (solid cross) and the SM prediction (triangle) are also indicated. A new physics scale of $\Lambda = 1$ TeV is assumed. Except for the two fitted Wilson coefficients, all others are set to zero. The CP-even c_{HW} vs $c_{H\overline{B}}$ scan uses the unfolded ($p_T^{4\ell}, m_{34}$) measurement, while the other scans use the unfolded ϕ measurement.

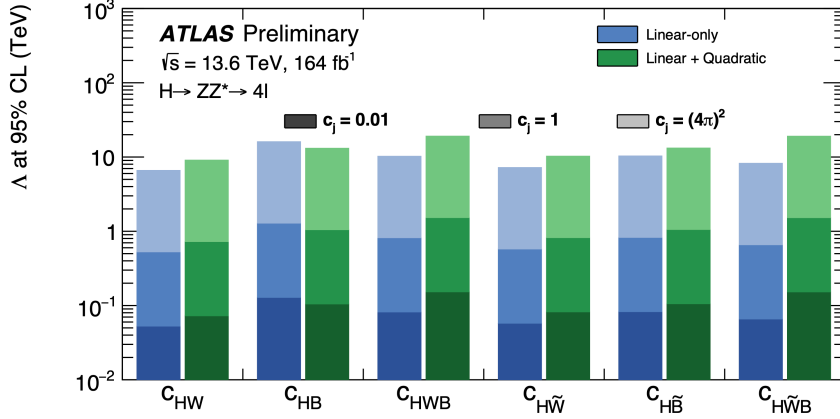


Figure 18: Observed lower limits at 95% confidence intervals (CL) on the new physics scale Λ for the indicated benchmark values of a given Wilson coefficient, assuming all others are set to zero. Asymmetric uncertainties are symmetrised using the root mean square. Based on the sensitivity current results, the constraints for $c_j = 0.01$ are outside the validity of the EFT approach.

9.3 Constraints on Higgs boson self-coupling

The trilinear Higgs self-coupling, λ_3 , enters single-Higgs production and decay at the one-loop level through electroweak corrections, and can therefore be constrained using precision measurements of single-Higgs observables. These effects modify both the overall rate and, in particular, the shape of kinematic distributions that probe hard radiation, making the Higgs boson transverse-momentum spectrum especially sensitive to λ_3 [139]. For this reason, both the differential measurement of $p_T^{4\ell}$ in the fiducial phase space and the STXS measurements, which rely on different event categorisations and modelling assumptions, can be used to probe the effects of the self-coupling, allowing a comparison of the sensitivity achieved by the two complementary approaches.

The interpretation is performed in the κ -framework by introducing the modifier $\kappa_\lambda \equiv \lambda_3/\lambda_3^{\text{SM}}$, with $\kappa_\lambda = 1$ corresponding to the SM expectation and setting all other couplings to their SM expectations.

The dependence of an observable Σ on the trilinear Higgs self-coupling is parametrised following Ref. [139] as

$$\Sigma_{\lambda_3}^{\text{BSM}}(\kappa_\lambda) = Z_H^{\text{BSM}} \Sigma_{\text{LO}} (1 + \kappa_\lambda C_1(\{p_n\}) + \delta Z_H), \quad (6)$$

where Σ_{LO} denotes the leading-order prediction, including factorisable higher-order QCD corrections. The factor Z_H^{BSM} represents the universal Higgs wave-function renormalisation, which accounts for corrections induced by the modified self-coupling. The coefficient $C_1(\{p_n\})$ encodes the finite one-loop contribution arising from the modified trilinear interaction and depends on the kinematics of the process, denoted collectively by $\{p_n\}$. The factor Z_H^{BSM} is expressed in terms of δZ_H and represents the NLO term [139], as

$$Z_H^{\text{BSM}} = \frac{1}{1 - (\kappa_\lambda^2 - 1) \delta Z_H}. \quad (7)$$

In this interpretation, all other coupling modifiers are fixed to their SM values, and the κ_λ dependence is implemented as a reweighting of the expected signal yields in each analysis bin, using C_1 values evaluated in the relevant fiducial or STXS phase space. The C_1 coefficients are obtained with the automated reweighting

procedure of Ref. [139] for all production-modes supported by the tool, and supplemented for ggF using the dedicated calculation at $\sqrt{s} = 13.6$ TeV [140]. The Higgs decay contribution is taken to be weakly dependent on kinematics [139], and the corresponding total-width contribution is taken from Ref. [141]. The range of the validity for the given C_1 and δZ_H is $|\kappa_\lambda| < 20$.

Complementary constraints on κ_λ are derived from the differential $p_T^{4\ell}$ and the STXS production-mode measurements. The corresponding likelihood scans are compared in Figure 19, yielding observed 95% CL intervals of $[-8.8, 19.9]$ and $[-9.0, 20.9]$, respectively, compatible with expected 95% CL intervals of $[-8.8, 19.9]$. The double-minimum structure of the likelihood arises from the quadratic dependence of the cross section on κ_λ . The observed limit from the STXS measurement is slightly weaker than that obtained from the differential $p_T^{4\ell}$ measurement. This difference is driven by the strong sensitivity of the κ_λ interpretation to the ggF production mode and by the fact that the observed ggF signal strength in the STXS measurement is lower than the inclusive signal strength.

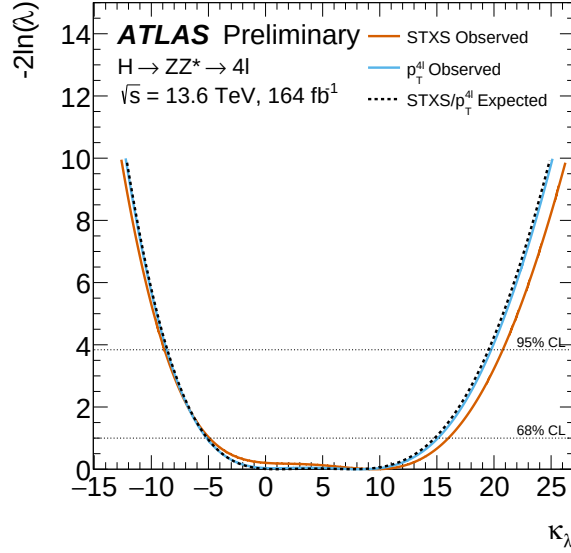


Figure 19: Observed negative-log-likelihood scan as a function of the trilinear Higgs self-coupling modifier κ_λ , comparing the constraints obtained from the STXS production-mode measurement (orange line) and from the fiducial differential measurement based on $p_T^{4\ell}$ (light blue line). The expected results are plotted together for measurements due to their similar sensitivity (black dashed line). The curves are shown relative to their respective minima. The Standard Model prediction corresponds to $\kappa_\lambda = 1$.

10 Conclusion

A measurement of inclusive and differential Higgs boson production cross-sections in the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel is presented using 164 fb^{-1} of $\sqrt{s} = 13.6$ TeV pp collision data recorded by the ATLAS detector at the LHC between 2022 and 2024. The inclusive fiducial cross-section is measured to be $\sigma_{\text{fid}} = 3.65^{+0.35}_{-0.33} \text{ fb}$, in agreement with the Standard Model prediction of $\sigma_{\text{fid}}^{\text{SM}} = 3.68 \pm 0.17 \text{ fb}$, and is extrapolated to a total cross-section of $\sigma_{\text{total}} = 59.3 \pm 5.7 \text{ pb}$, also consistent with the Standard Model and previous results. Differential cross-sections are measured as a function of several Higgs boson production and decay observables and are found to be in agreement with state-of-the-art SM predictions. Higgs boson candidate events are categorised

into regions sensitive to different production mechanisms and are used to measure production-mode cross-sections and a Reduced STXS Stage 1.2 scheme. A product of inclusive production cross-section and branching ratio into ZZ^* of $\sigma_{|y_H|<2.5} \cdot \mathcal{B}(H \rightarrow ZZ^*) = 1.42 \pm 0.16$ pb is obtained for Higgs boson rapidities below 2.5, consistent with the Standard Model prediction of $\sigma_{|y_H|<2.5}^{\text{SM}} \cdot \mathcal{B}^{\text{SM}}(H \rightarrow ZZ^*) = 1.44 \pm 0.08$ pb. The precision of both the differential and STXS measurements is comparable to that achieved by ATLAS at $\sqrt{s} = 13$ TeV with Run 2 data [8, 9], and improves by approximately a factor of two relative to the first ATLAS results at $\sqrt{s} = 13.6$ TeV in this channel [20]. The differential measurements as a function of ϕ and $p_T^{4\ell}$ vs. m_{34} are used to constrain effective field theory operators and are found to be compatible with SM expectations. The production-mode results are further interpreted in the κ framework and combined with Run 2 measurements [8], yielding best-fit values consistent with Run 2 and about 40% (25%) improvement on κ_V (κ_f) limits. Limits are also derived on the trilinear Higgs self-coupling, and are consistent with the Standard Model.

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